APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – I

Introduction and Solution Methods

Syllabus (as given)

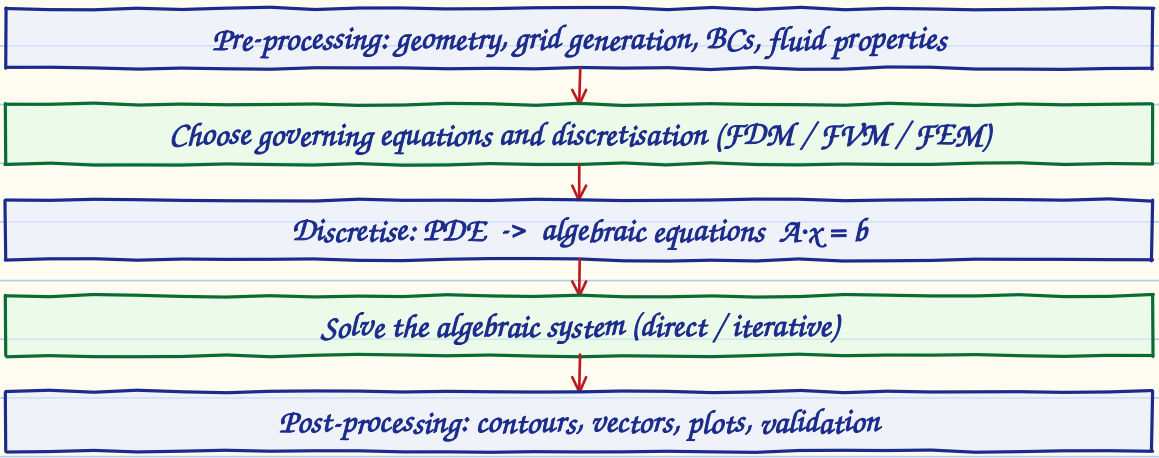
- Introduction: finite difference, finite volume and finite element methods, governing equations and boundary conditions, derivation of finite difference equations.
- Elliptic equations: finite difference formulations, iterative solution methods, direct method with Gaussian elimination.
- Parabolic equations: explicit schemes and Von Neumann stability analysis, implicit schemes, ADI, approximate factorization, fractional step methods, direct method with tridiagonal matrix algorithm (TDMA).

1. What is CFD?

Computational Fluid Dynamics = analysing fluid flow, heat transfer and related phenomena (combustion, chemical reactions) by solving the governing equations numerically on a computer. The continuous problem (PDEs) is converted into a discrete algebraic problem.

- Three tools of fluid dynamics: theory (analytical), experiment (wind tunnel), and CFD (numerical).
- Advantages: low cost, fast design cycles, full flow-field details, can simulate conditions that are impossible/dangerous to test.
- Limitations: needs modelling (turbulence, chemistry), numerical errors, results must be validated.

Steps in a CFD analysis



2. Governing Equations

All fluid-flow equations come from three conservation laws: conservation of mass, momentum (Newton's 2nd law) and energy (1st law of thermodynamics).

Continuity (mass)

$$\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0 \quad \text{i.e.} \quad \partial\rho/\partial t + \nabla\cdot(\rho V) = 0$$

For incompressible flow (ρ = constant): $\nabla\cdot V = 0$.

Momentum (Navier–Stokes), x-direction

$$\rho(\partial u/\partial t + u\partial u/\partial x + v\partial u/\partial y + w\partial u/\partial z) = -\partial p/\partial x + \mu\nabla^2 u + \rho f_x$$

Similar equations for y and z. Left side = inertia (unsteady + convective); right side = pressure force + viscous force + body force.

Energy

$$\rho c_p(\partial T/\partial t + V\cdot\nabla T) = k\nabla^2 T + \Phi + q$$

Φ = viscous dissipation, q = heat source. For a solid, or low-speed flow, this reduces to the heat equation $\partial T/\partial t = \alpha\nabla^2 T$ with $\alpha = k/\rho c_p$.

Generic transport equation (very important)

$$\partial\varphi/\partial t + \nabla\cdot(\rho V\varphi) = \nabla\cdot(\Gamma\nabla\varphi) + S_\varphi$$

- terms: unsteady + convection = diffusion + source
- $\varphi = 1$ gives continuity, $\varphi = u$ gives momentum, $\varphi = T$ gives energy. One discretisation method serves all equations.

3. Classification of PDEs

Second-order quasi-linear PDE: $A\varphi_{xx} + B\varphi_{xy} + C\varphi_{yy} + D = 0$. Type depends on the discriminant $B^2 - 4AC$.

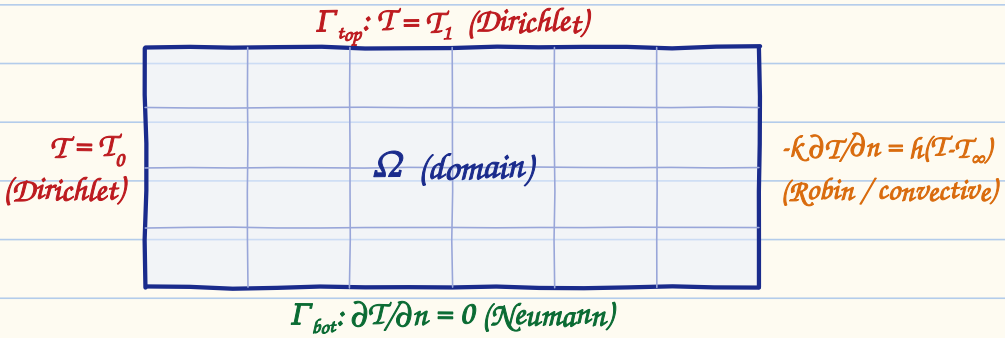
Type	B^2-4AC	Example	Physics
Elliptic	< 0	Laplace $\nabla^2\varphi=0$	steady, equilibrium
Parabolic	$= 0$	Heat $u_t=\alpha u_{xx}$	diffusion in time
Hyperbolic	> 0	Wave $u_{tt}=c^2u_{xx}$	wave propagation

- Elliptic: every point influences every other point; boundary conditions needed on the whole closed boundary -> boundary-value problem; solve all nodes together.
- Parabolic: information travels forward in time, spreads in space -> marching problem (initial +

boundary conditions).

- Hyperbolic: finite speed of propagation along characteristics -> marching with domain of dependence.

4. Boundary Conditions



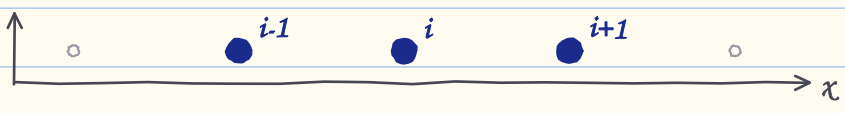
Type	Specified	Example
Dirichlet (1st)	value of φ	wall temperature, no-slip $u = 0$
Neumann (2nd)	normal derivative $\partial\varphi/\partial n$	insulated wall, symmetry plane
Robin (3rd)	$a\varphi + b\,\partial\varphi/\partial n = c$	convection $-kT_n = h(T - T_\infty)$

- A problem is well-posed if solution exists, is unique and depends continuously on data (Hadamard).
- Neumann conditions are implemented with a fictitious (ghost) node: $\partial\varphi/\partial x|_0 = 0$ -> $\varphi_{-1} = \varphi_1$.
- Initial condition $\varphi(x,0) = f(x)$ is needed for unsteady (parabolic / hyperbolic) problems.

5. Finite Difference Method (FDM)

Idea: replace the domain by a grid of discrete points and replace each derivative in the PDE by an algebraic difference of nodal values. The oldest and simplest method, best on structured grids.

1-D grid, spacing Δx



Derivation using Taylor series

$$u_{i+1} = u_i + \Delta x\, u_x + (\Delta x^2/2)\, u_{xx} + (\Delta x^3/6)\, u_{xxx} + (\Delta x^4/24)\, u_{xxxx} + \dots \quad (1)$$

$$u_{i-1} = u_i - \Delta x\, u_x + (\Delta x^2/2)\, u_{xx} - (\Delta x^3/6)\, u_{xxx} + (\Delta x^4/24)\, u_{xxxx} - \dots \quad (2)$$

(a) From (1): $u_x = (u_{i+1} - u_i)/\Delta x - (\Delta x/2)u_{xx} \dots$ Forward difference, $O(\Delta x)$

(b) From (2): $u_x = (u_i - u_{i-1})/\Delta x + (\Delta x/2)u_{xx} \dots$ Backward difference, $O(\Delta x)$

(c) (1) - (2): $u_x = (u_{i+1} - u_{i-1})/2\Delta x - (\Delta x^2/6)u_{xxx}$ Central difference, $O(\Delta x^2)$

(d) (1) + (2): $u_{xx} = (u_{i+1} - 2u_i + u_{i-1})/\Delta x^2 - (\Delta x^2/12)u_{xxxx}$ $O(\Delta x^2)$

The terms dropped are the truncation error; its leading power of Δx gives the order of accuracy.

Derivative	Formula	Order
u_x forward	$(u_{i+1} - u_i)/\Delta x$	1
u_x backward	$(u_i - u_{i-1})/\Delta x$	1
u_x central	$(u_{i+1} - u_{i-1})/(2\Delta x)$	2
u_{xx} central	$(u_{i+1} - 2u_i + u_{i-1})/\Delta x^2$	2
u_x 2nd-order upwind	$(3u_i - 4u_{i-1} + u_{i-2})/2\Delta x$	2

Mixed derivative: $u_{xy} \approx (u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}) / (4\Delta x\Delta y)$.

Concepts to remember

- Consistency: truncation error $\to 0$ as $\Delta x, \Delta t \to 0$.
- Stability: errors (round-off) do not grow as the solution marches.
- Convergence: numerical solution $\to$ exact solution as grid is refined.
- Lax equivalence theorem: for a well-posed linear initial-value problem, a consistent scheme is convergent if and only if it is stable.

6. Finite Volume and Finite Element Methods (overview)

Finite Volume Method (FVM)

- Domain divided into control volumes; the conservation law is integrated over each CV and the divergence theorem converts volume integrals into face fluxes.
- Fluxes leaving one cell = fluxes entering the neighbour -> conservative by construction. Works on unstructured grids. Basis of most commercial codes (Fluent, OpenFOAM, CFX).

Finite Element Method (FEM)

- Domain divided into elements (triangles / quadrilaterals); solution approximated as $u \approx \Sigma N_j(x)u_j$ with shape functions N_j; weighted-residual (Galerkin) or variational statement gives element matrices, then assembly.
- Excellent for complex geometry and natural boundary conditions; strong in structural and

Stokes-type flows (see Unit V).

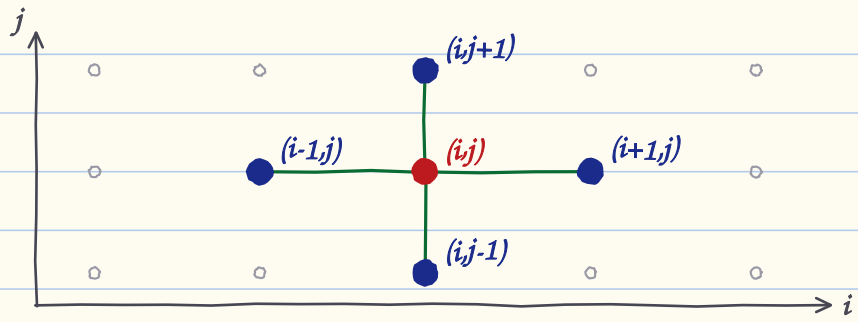
Feature	FDM	FVM	FEM
Basis	Taylor series	Integral balance	Weighted residual
Grid	structured	any	any (unstructured)
Conservation	not guaranteed	exact	weak sense
Complex geometry	difficult	good	very good

7. Solution Methods for Elliptic Equations

Model problem: Poisson equation $\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = S(x,y)$ on a rectangle with φ given on the boundary (Dirichlet).

Finite-difference formulation (five-point formula)

Five-point stencil



Central differences in both directions with Δx, Δy:

$$(\varphi_{i+1,j} - 2\varphi_{i,j} + \varphi_{i-1,j})/\Delta x^2 + (\varphi_{i,j+1} - 2\varphi_{i,j} + \varphi_{i,j-1})/\Delta y^2 = S_{i,j}$$

With $\Delta x = \Delta y = h$ this gives the famous five-point formula:

$$\varphi_{i,j} = \tfrac{1}{4}\,(\varphi_{i+1,j} + \varphi_{i-1,j} + \varphi_{i,j+1} + \varphi_{i,j-1}) - (h^2/4)\,S_{i,j}$$

- Each interior node gives one linear equation; N interior nodes -> N equations -> system $A\varphi = b$ with a sparse, banded, diagonally dominant matrix.
- Value at a node = average of its four neighbours (Laplace equation) -> maximum principle: no interior maxima/minima.

8. Iterative (Interactive) Solution Methods

For large grids the matrix is huge but sparse; direct inversion is costly, so we start with a guess and improve it repeatedly. Write the five-point formula in iteration form ($S = 0$, $\Delta x = \Delta y$):

(i) Jacobi method

$$\varphi_{ij}^{(k+1)} = \tfrac{1}{4}\left(\varphi_{i+1,j}^{(k)} + \varphi_{i-1,j}^{(k)} + \varphi_{i,j+1}^{(k)} + \varphi_{i,j-1}^{(k)}\right)$$

Uses only old values; needs two arrays; slow.

(ii) Gauss–Seidel method

$$\varphi_{ij}^{(k+1)} = \tfrac{1}{4}\left(\varphi_{i+1,j}^{(k)} + \varphi_{i-1,j}^{(k+1)} + \varphi_{i,j+1}^{(k)} + \varphi_{i,j-1}^{(k+1)}\right)$$

Uses latest available values (sweeping i, j increasing); about twice as fast as Jacobi; one array only.

(iii) Successive Over-Relaxation (SOR)

$$\varphi^{(k+1)} = (1-\omega)\,\varphi^{(k)} + \omega\,\varphi_{GS}^{(k+1)}, \qquad 1 < \omega < 2$$

- Optimum factor for a square grid ($N \times N$): $\omega_{opt} \approx 2 / (1 + \sin(\pi/N))$.
- Line SOR / line Gauss–Seidel: solve a whole grid line implicitly with TDMA, sweep line by line (faster convergence).
- Convergence criterion: $\max |\varphi^{(k+1)} - \varphi^{(k)}| < \varepsilon$ (e.g. 10^{-5}), or residual norm $||b - A\varphi|| < \varepsilon$.
- Gauss–Seidel / Jacobi converge when the matrix is diagonally dominant (Scarborough: $\Sigma |a_{nb}| \leq |a_p|$, strict at least once).

Method	Iterations	Storage	Remark
Jacobi	~ N^2	2 arrays	parallel friendly
Gauss–Seidel	~ $N^2/2$	1 array	2× Jacobi
SOR (ω opt)	~ N	1 array	fast, ω critical

Worked example: Gauss–Seidel on a 2×2 interior grid

Plate: top edge = 100°, other three edges = 0°. Unknowns A, B (top row), C, D (bottom row).

$A = (100 + B + C)/4$, $B = (100 + A + D)/4$, $C = (A + D)/4$, $D = (B + C)/4$. Start $A = B = C = D = 0$.

Iteration 1: $A = 25$, $B = 31.25$, $C = 6.25$, $D = 9.375$

Iteration 2: $A = 34.375$, $B = 35.94$, $C = 10.94$, $D = 11.72$

Exact (by symmetry): $A = B = 37.5$, $C = D = 12.5$ -> converging.

9. Direct Method: Gaussian Elimination

Solves $A x = b$ exactly (up to round-off) in a finite number of operations.

1. Forward elimination: use row k to eliminate x_k from all rows below: $a_{ij} \leftarrow a_{ij} - (a_{ik}/a_{kk})\,a_{kj}$, $b_i \leftarrow$

$b_i - (a_{ik}/a_{kk})\, b_k$. The matrix becomes upper triangular $U\,x = c$.

2. **Back substitution**: $x_n = c_n/u_{nn}$, $x_i = (c_i - \Sigma_{j>i}\, u_{ij}x_j)/u_{ii}$.
3. Partial pivoting (swap rows so that the pivot is largest) prevents division by a tiny number.

- Operation count ≈ $(2/3)n^3$ for full matrix; for a banded matrix of half-bandwidth m only ≈ $n\,m^2$. For an $N\times N$ grid, $n = N^2$, $m = N$ -> N^4 operations — expensive, so iterative / fast solvers preferred for big 2-D, 3-D problems.
- Disadvantages: fill-in inside the band, large memory, round-off accumulation.

Remember

- Direct: exact, robust, cost grows rapidly with n.
- Iterative: cheap per iteration, needs convergence; best for large sparse systems.

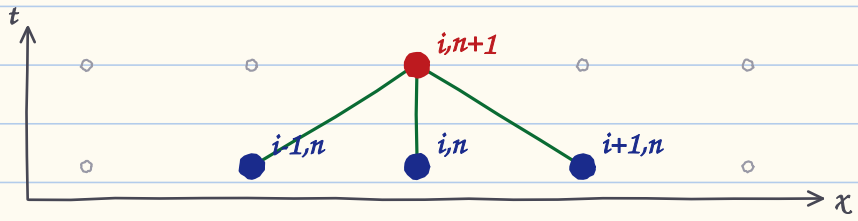
10. Parabolic Equations

Model: one-dimensional unsteady heat (diffusion) equation $\partial u/\partial t = \alpha\, \partial^2 u/\partial x^2$, $0 < x < L$, $t > 0$, with initial $u(x,0)$ and boundary values $u(0,t)$, $u(L,t)$.

Grid: $x_i = i\Delta x$, $t^n = n\Delta t$, $u_i^n \approx u(x_i, t^n)$. Diffusion number $r = \alpha\Delta t/\Delta x^2$.

Explicit scheme: FTCS (Forward-Time, Central-Space)

FTCS stencil: one unknown



$$(u_i^{n+1} - u_i^n)/\Delta t = \alpha\, (u_{i+1}^n - 2u_i^n + u_{i-1}^n)/\Delta x^2$$

$$u_i^{n+1} = r\, u_{i-1}^n + (1 - 2r)\, u_i^n + r\, u_{i+1}^n$$

- Truncation error $O(\Delta t) + O(\Delta x^2)$; each new value is found directly (no matrix) — cheap and simple.
- Condition for positive coefficients (physically realistic, no oscillations): $r \le ½$.

11. Von Neumann Stability Analysis

Method: represent the error (or solution) as a Fourier mode and see whether its amplitude grows or decays in one time step. Valid for linear, constant-coefficient schemes (periodic / ignoring boundaries).

Step 1 Substitute $u_i^n = g^n e^{I k x_i} = g^n e^{I i\theta}$, $\theta = k\Delta x$ ($I = \sqrt{-1}$).

Step 2 Cancel common factors -> amplification factor $G = g^{n+1}/g^n$.

Step 3 Stable if $|G| \le 1$ for all θ in $[0, \pi]$.

Apply to FTCS heat equation

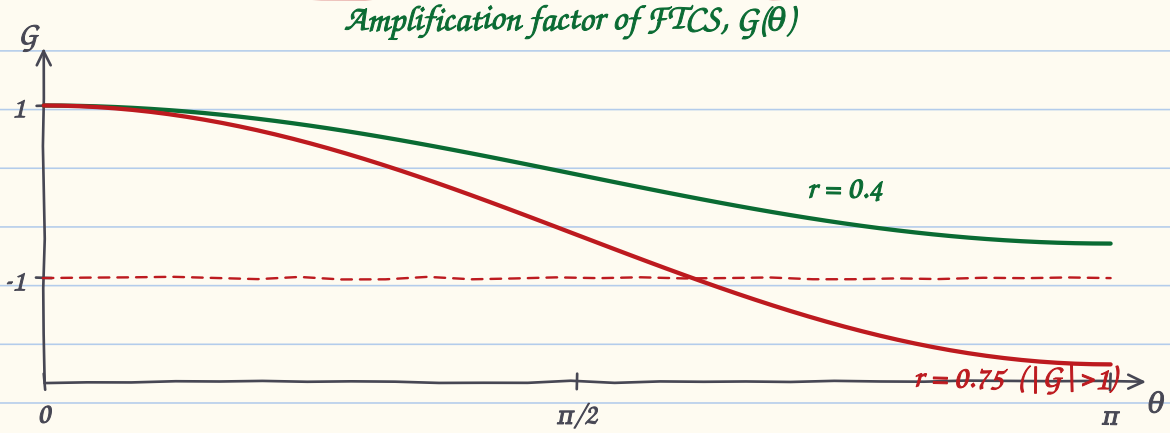
$$G\, e^{I\theta} = r\, e^{I(i-1)\theta} + (1-2r)\, e^{I\theta} + r\, e^{I(i+1)\theta}$$

$$G = 1 - 2r + 2r\cos\theta = 1 - 4r\sin^2(\theta/2)$$

G is real, $G \le 1$ always. Need $G \ge -1$: $1 - 4r \ge -1$ (worst case $\theta = \pi$) ->

$$r = \alpha\Delta t/\Delta x^2 \le 1/2 \quad \text{(conditionally stable)}$$

Amplification factor of FTCS, $G(\theta)$

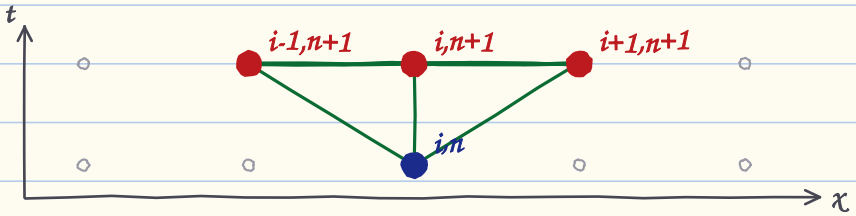


- Time step restriction $\Delta t \le \Delta x^2/(2\alpha)$: halving Δx forces Δt to be 4 times smaller -> explicit is costly on fine grids.

12. Implicit Schemes

Fully implicit (BTCS, Laasonen)

BTCS stencil: three unknowns



$$-r\, u_{i-1}^{n+1} + (1+2r)\, u_i^{n+1} - r\, u_{i+1}^{n+1} = u_i^n$$

- Gives a tridiagonal system at every time level (solved by TDMA). Von Neumann: $G = 1/(1 + 4r\sin^2(\theta/2))$ -> $|G| \le 1$ for all r: unconditionally stable.
- Accuracy $O(\Delta t) + O(\Delta x^2)$; Δt can be large, limited only by accuracy.

Crank–Nicolson (average of n and n+1)

$$-(r/2)u_{i-1}^{n+1} + (1+r)u_i^{n+1} - (r/2)u_{i+1}^{n+1} = (r/2)u_{i-1}^{n} + (1-r)u_i^{n} + (r/2)u_{i+1}^{n}$$

- Second order in time and space $O(\Delta t^2) + O(\Delta x^2)$; $G = (1 - 2r \sin^2(\theta/2))/(1 + 2r \sin^2(\theta/2))$; unconditionally stable; may oscillate for very large r.

Scheme	Order	Stability	Matrix
FTCS	$\Delta t, \Delta x^2$	$r \le 1/2$	none
BTCS	$\Delta t, \Delta x^2$	unconditional	tridiagonal
Crank–Nicolson	$\Delta t^2, \Delta x^2$	unconditional	tridiagonal
Richardson	$\Delta t^2, \Delta x^2$	unstable always	none
DuFort–Frankel	$\Delta t^2, \Delta x^2$	unconditional	none

Note: Richardson (leapfrog in time) is always unstable for diffusion; DuFort–Frankel replaces $2u_i^n$ by $(u_i^{n+1} + u_i^{n-1})$ and becomes stable but may be inconsistent unless $\Delta t/\Delta x \to 0$.

13. Direct Method: Tridiagonal Matrix Algorithm (Thomas)

Tridiagonal system $A \cdot T = d$

$$\begin{vmatrix} b_1 & c_1 & 0 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 0 \\ 0 & a_3 & b_3 & c_3 & 0 \\ 0 & 0 & a_4 & b_4 & c_4 \\ 0 & 0 & 0 & a_5 & b_5 \end{vmatrix} \quad [T_1\ T_2\ T_3\ T_4\ T_5] = [d_1\ d_2\ d_3\ d_4\ d_5]$$

b = diagonal, c = upper, a = lower

Equation i: $a_i T_{i-1} + b_i T_i + c_i T_{i+1} = d_i$. Gaussian elimination reduces to this simple two-sweep algorithm:

1. Forward sweep: $c'_1 = c_1/b_1$, $d'_1 = d_1/b_1$;

$$c'_i = c_i/(b_i - a_i c'_{i-1}), \quad d'_i = (d_i - a_i d'_{i-1})/(b_i - a_i c'_{i-1})$$

2. Back substitution: $T_n = d'_n$, $T_i = d'_i - c'_i T_{i+1}$ $(i = n-1, \ldots, 1)$.

- Only about 8n operations — O(n) instead of $O(n^3)$; needs diagonal dominance ($|b_i| > |a_i| + |c_i|$) for stability without pivoting.

Worked example

$2T_1 - T_2 = 1$; $-T_1 + 2T_2 - T_3 = 0$; $-T_2 + 2T_3 = 1$

$c'_1 = -1/2$, $d'_1 = 1/2$; $c'_2 = -1/(2 - 1/2) = -2/3$, $d'_2 = (0 + 1/2)/(3/2) = 1/3$; $d'_3 = (1 + 1/3)/(2 - 2/3) = 1$

Back substitution: $T_3 = 1$, $T_2 = 1/3 + (2/3)(1) = 1$, $T_1 = 1/2 + (1/2)(1) = 1$. Check: $2(1) - 1 = 1$ √

14. Multi-dimensional Parabolic Problems

2-D heat equation $u_t = \alpha(u_{xx} + u_{yy})$. Explicit FTCS needs $r_x + r_y \le ½$ (for $\Delta x = \Delta y$: $r \le ¼$). Fully implicit gives a pentadiagonal (5-point) matrix at each step — TDMA no longer applies directly.

Alternating Direction Implicit (ADI) — Peaceman–Rachford

Split each time step into two half steps; implicit in one direction, explicit in the other, so that only tridiagonal systems arise.

$$\text{Half step 1:}\quad (1 - (r_x/2)\delta_x^2)\, u^{n+1/2} = (1 + (r_y/2)\delta_y^2)\, u^n \qquad \text{(implicit in } x\text{)}$$

$$\text{Half step 2:}\quad (1 - (r_y/2)\delta_y^2)\, u^{n+1} = (1 + (r_x/2)\delta_x^2)\, u^{n+1/2} \qquad \text{(implicit in } y\text{)}$$

- $\delta_x^2 u = u_{i+1} - 2u_i + u_{i-1}$ (central second difference operator).
- Each half step = set of independent tridiagonal systems along grid lines -> TDMA. Second order $O(\Delta t^2, \Delta x^2)$, unconditionally stable in 2-D (not in 3-D with the simple extension; use Douglas–Gunn).

Approximate Factorization (Beam–Warming, delta form)

Implicit 2-D scheme: $(1 - r\theta(\delta_x^2 + \delta_y^2))\,\Delta u = r(\delta_x^2 + \delta_y^2)u^n$, $\Delta u = u^{n+1} - u^n$. Factor the operator:

$$(1 - r\theta\delta_x^2)(1 - r\theta\delta_y^2)\,\Delta u = r(\delta_x^2 + \delta_y^2)\,u^n$$

- The product introduces an extra term $r^2\theta^2\delta_x^2\delta_y^2\Delta u = O(\Delta t^3)$ — same order as the truncation error, so accuracy is preserved.
- Solve two sweeps: $(1 - r\theta\delta_x^2)\Delta u^* = RHS$ then $(1 - r\theta\delta_y^2)\Delta u = \Delta u^*$, each by TDMA; update $u^{n+1} = u^n + \Delta u$.

Fractional Step (Yanenko) Method

Split the multi-dimensional operator into 1-D operators and advance them successively:

$$(1 - r\delta_x^2)\, u^* = u^n \;; \qquad (1 - r\delta_y^2)\, u^{n+1} = u^*$$

- Each step is a 1-D implicit solve (TDMA). First order in time in general (split error); stable for any r. Easily extended to 3-D and to non-linear terms (each fractional step handles one physical

process: convection, diffusion, source).

15. Worked Example: Explicit Heat Conduction

A rod of length 1 m, $\alpha = 0.01$ m²/s, initial temperature 0, ends held at $u(0,t) = 0$ and $u(1,t) = 100$. Take $\Delta x = 0.25$ m (3 interior nodes) and $\Delta t = 1$ s.

1. Diffusion number $r = \alpha\Delta t/\Delta x^2 = 0.01(1)/0.0625 = 0.16 \le 0.5$ -> explicit scheme is stable.
2. Update formula: $u_i^{n+1} = 0.16\, u_{i-1}^n + 0.68\, u_i^n + 0.16\, u_{i+1}^n$.
3. Step 1 ($n = 0$ -> 1): $u_1 = 0.16(0) + 0.68(0) + 0.16(0) = 0$, $u_2 = 0$, $u_3 = 0.16(0) + 0.68(0) + 0.16(100) = 16$.
4. Step 2: $u_1 = 0$, $u_2 = 0.16(0) + 0.68(0) + 0.16(16) = 2.56$, $u_3 = 0.16(0) + 0.68(16) + 0.16(100) = 26.88$.

Heat diffuses in from the hot end one node per time step — a visible sign that explicit schemes carry information over only one cell per step.

16. Truncation Error, Modified Equation and Accuracy

Substituting Taylor expansions into FTCS gives the modified equation that the scheme really solves:

$$u_t = \alpha\, u_{xx} + (\Delta t/2)\, u_{tt} - (\alpha\Delta x^2/12)\, u_{xxxx} + \dots$$

- Leading error terms: $O(\Delta t)$ from the forward time difference and $O(\Delta x^2)$ from the central space difference.
- Choosing $r = 1/6$ ($\Delta t = \Delta x^2/6\alpha$) cancels the two leading terms and gives $O(\Delta t^2, \Delta x^4)$ accuracy for the explicit scheme — a useful trick.
- Grid refinement test: error ratio between grids Δx and $\Delta x/2$ is $\approx 2^p$; this gives the observed order p.
- Round-off errors are amplified when $|G| > 1$; discretisation errors are controlled by order + grid size. Stability is separate from accuracy.

Boundary treatment in the heat equation

- Dirichlet: simply use the given value for u_0^n and u_N^n.
- Neumann $\partial u/\partial x = q$ at $x = 0$: one-sided $u_x \approx (-3u_0 + 4u_1 - u_2)/2\Delta x = q$ (2nd order), or ghost node $u_{-1} = u_1 - 2\Delta x\, q$ (central).
- Insulated end ($q = 0$): $u_{-1} = u_1$, so $u_0^{n+1} = 2r\, u_1^n + (1-2r)\, u_0^n$.

Quick revision — Unit I

- Parabolic explicit FTCS: $r = \alpha\Delta t/\Delta x^2 \le 1/2$; BTCS and Crank–Nicolson: unconditionally stable.
- Von Neumann: put $u = g^n e^{I\theta}$, find G, need $|G| \le 1$.
- Elliptic: five-point formula; Jacobi < Gauss–Seidel < SOR.
- Direct: Gauss elimination $O(n^3)$; TDMA $O(n)$ for tridiagonal systems.
- Multi-D implicit: ADI, approximate factorization, fractional step — all reduce to repeated TDMA.

Important Questions

1. Derive central, forward and backward difference formulas using Taylor series and state their order.
2. Classify PDEs and give one example for each class.
3. Derive the five-point formula; compare Jacobi, Gauss–Seidel and SOR.
4. Explain Von Neumann stability analysis and find the stability limit of FTCS for the heat equation.
5. Explain Thomas algorithm with an example; explain ADI and fractional step methods.

APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – II

Hyperbolic Equations

Syllabus (as given)

- Hyperbolic equations: explicit schemes and Von Neumann stability analysis, implicit schemes, multi-step methods, nonlinear problems, second-order one-dimensional wave equations.
- Burgers equations: explicit and implicit schemes, Runge–Kutta method.

1. Hyperbolic Equations — Introduction

Hyperbolic PDEs describe wave propagation and convection (transport without diffusion): sound waves, shock waves, supersonic flow, flood waves. Two basic model equations:

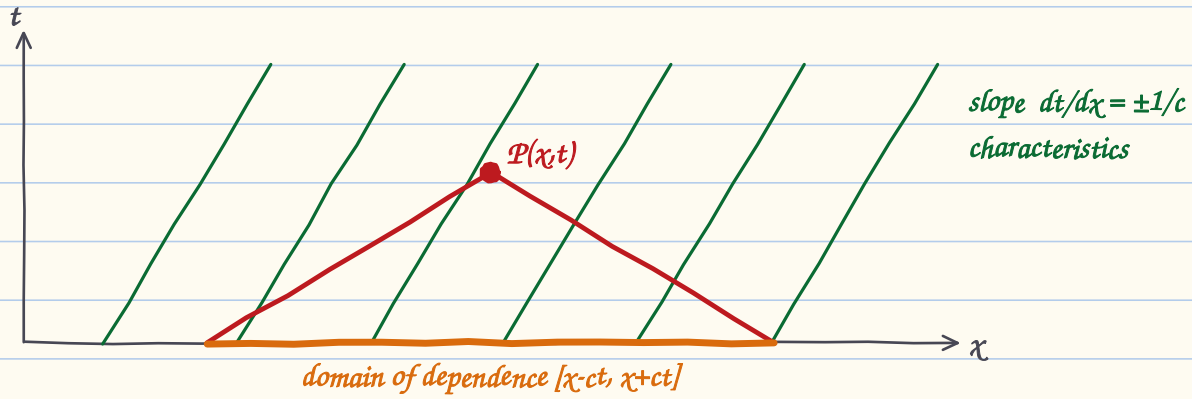
$$\text{First-order wave (linear convection):}\quad \partial u/\partial t + c\,\partial u/\partial x = 0\,,\quad c > 0$$

$$\text{Second-order wave equation:}\quad \partial^2 u/\partial t^2 = c^2\,\partial^2 u/\partial x^2$$

Exact solution of the first-order equation: $u(x,t) = f(x - ct)$ — the initial profile $f(x)$ simply translates to the right with speed c without changing shape.

Characteristics and domain of dependence

Along the lines $dx/dt = c$ (characteristics) the solution u is constant. Information travels at finite speed c along these lines.



- Value at $P(x,t)$ depends only on the initial data in the domain of dependence $[x - ct, x + ct]$ (for the 2nd-order equation); the region influenced by a point is its range of influence.
- A numerical scheme must respect this: its numerical domain of dependence must contain the

physical one -> CFL condition.

Courant–Friedrichs–Lewy (CFL) condition

$$\nu = c\,\Delta t / \Delta x \le 1$$

- ν is the Courant number (CFL number). It means that the wave must not travel more than one grid cell in one time step in an explicit scheme.
- CFL is necessary for stability of explicit schemes, but not always sufficient.

2. Explicit Schemes for $u_t + c\,u_x = 0$

Notation: $u_i^n = u(i\Delta x, n\Delta t)$, $\nu = c\Delta t/\Delta x$, Von Neumann: $u_i^n = g^n e^{I\theta}$, $G = g^{n+1}/g^n$.

(a) FTCS — forward time, central space

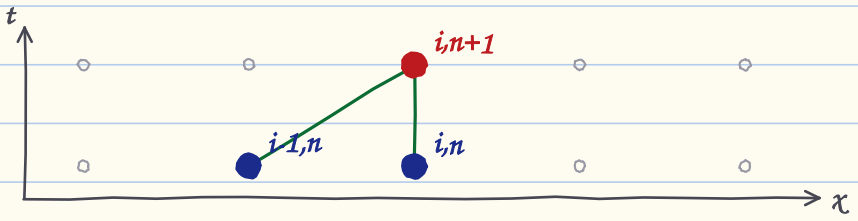
$$u_i^{n+1} = u_i^n - (\nu/2)(u_{i+1}^n - u_{i-1}^n)$$

$$G = 1 - (\nu/2)(e^{I\theta} - e^{-I\theta}) = 1 - I\nu \sin\theta \quad \rightarrow \quad |G|^2 = 1 + \nu^2 \sin^2\theta > 1$$

Unconditionally unstable for every ν — never use FTCS for pure convection.

(b) First-order upwind (FTBS for c > 0)

Upwind stencil (c > 0)



$$u_i^{n+1} = u_i^n - \nu\,(u_i^n - u_{i-1}^n)$$

$$G = 1 - \nu(1 - e^{-I\theta}) \quad \rightarrow \quad |G|^2 = 1 - 2\nu(1-\nu)(1 - \cos\theta)$$

- Stable when $0 \le \nu \le 1$. Accuracy $O(\Delta t, \Delta x)$. Strongly diffusive: modified equation $u_t + c\,u_x = (c\Delta x/2)(1-\nu)\,u_{xx}$ (numerical / artificial viscosity smears sharp fronts).
- Physical reason: information comes from upstream, so use upstream points. If $c < 0$ use $u_{i+1} - u_i$.

(c) Lax (Lax–Friedrichs) scheme

$$u_i^{n+1} = \tfrac{1}{2}(u_{i+1}^n + u_{i-1}^n) - (\nu/2)(u_{i+1}^n - u_{i-1}^n)$$

$$G = \cos\theta - I\nu \sin\theta \quad \rightarrow \quad |G|^2 = \cos^2\theta + \nu^2 \sin^2\theta \le 1 \text{ for } \nu \le 1$$

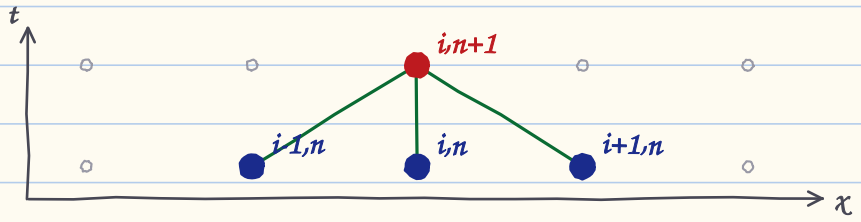
- Replaces u_i^n in FTCS by its spatial average -> stable ($\nu \le 1$), first-order, heavily dissipative.

(d) Lax–Wendroff scheme (second order)

Taylor series in time: $u^{n+1} = u^n + \Delta t\, u_t + (\Delta t^2/2)\, u_{tt}$, with $u_t = -c\, u_x$ and $u_{tt} = c^2 u_{xx}$. Central differences give:

$$u_i^{n+1} = u_i^n - (\nu/2)(u_{i+1}^n - u_{i-1}^n) + (\nu^2/2)(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

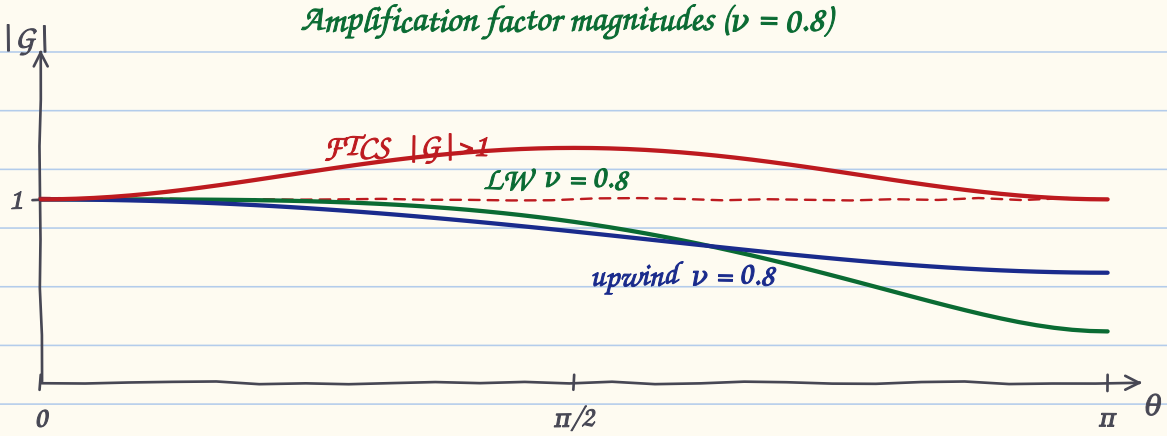
Lax–Wendroff stencil (3 known, 1 unknown)



$$G = 1 - \nu^2(1 - \cos\theta) - I\nu \sin\theta \quad \rightarrow \quad |G|^2 = 1 - 4\nu^2(1 - \nu^2)\sin^4(\theta/2)$$

- Stable for $\nu \le 1$; accuracy $O(\Delta t^2, \Delta x^2)$; produces dispersive errors (oscillations near discontinuities) rather than smearing.

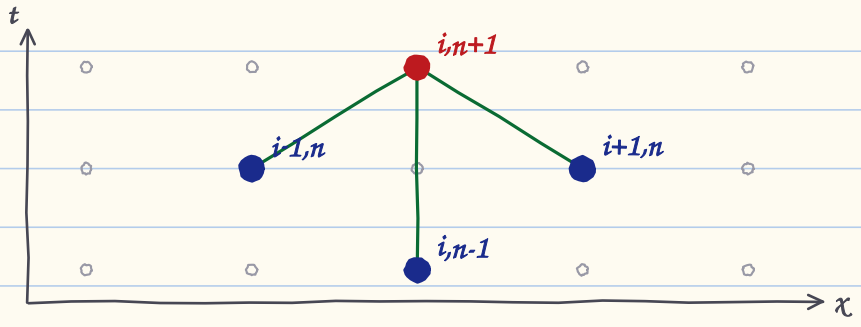
Amplification factor magnitudes ($\nu = 0.8$)



(e) Leapfrog (central in time and space)

$$u_i^{n+1} = u_i^{n-1} - \nu\,(u_{i+1}^n - u_{i-1}^n)$$

Leapfrog stencil (three time levels)



$$G^2 + 2I\nu \sin\theta \cdot G - 1 = 0 \;\rightarrow\; |G| = 1 \text{ for } \nu \le 1 \text{ (no dissipation)}$$

- Second order, non-dissipative, but needs two time levels (starter by another scheme) and has a spurious computational mode ($G_2 \approx -G_1$) causing odd–even decoupling.

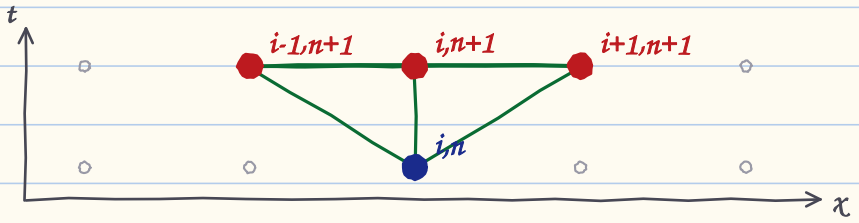
Scheme	Order	Stability	Main error
FTCS	$\Delta t, \Delta x^2$	never	growth
Upwind	$\Delta t, \Delta x$	$\nu \le 1$	diffusion
Lax	$\Delta t, \Delta x^2/\Delta t$	$\nu \le 1$	large diffusion
Lax–Wendroff	$\Delta t^2, \Delta x^2$	$\nu \le 1$	dispersion
Leapfrog	$\Delta t^2, \Delta x^2$	$\nu \le 1$	dispersion, 2 modes

3. Implicit Schemes

BTCS (fully implicit, central space)

$$u_i^{n+1} + (\nu/2)(u_{i+1}^{n+1} - u_{i-1}^{n+1}) = u_i^n$$

BTCS stencil for convection



$$G = 1/(1 + I\nu \sin\theta) \;\rightarrow\; |G|^2 = 1/(1 + \nu^2 \sin^2\theta) \le 1$$

- Unconditionally stable for any ν; tridiagonal system $(-\nu/2, 1, \nu/2)$ solved by TDMA each step; first order in time; strongly damped for large ν.

Crank–Nicolson

$$u_i^{n+1} - u_i^n = -(\nu/4)[(u_{i+1}^{n+1} - u_{i-1}^{n+1}) + (u_{i+1}^n - u_{i-1}^n)]$$

$G = (1 - I\nu \sin\theta/2)/(1 + I\nu \sin\theta/2) \;\rightarrow\; |G| = 1$ (no damping, unconditionally stable, $O(\Delta t^2, \Delta x^2)$)

Implicit upwind

$$(1 + \nu)\, u_i^{n+1} - \nu\, u_{i-1}^{n+1} = u_i^n \;\rightarrow\; \text{bidiagonal; stable for all } \nu$$

- Implicit methods remove the CFL restriction, but the wave is then not resolved accurately for large ν (phase errors). Cost: one tridiagonal solve per step. For nonlinear systems the implicit equations must be linearised.

4. Multi-step Methods

Advance the solution in two (or more) stages per time step to get high order without computing high derivatives directly.

Two-step Lax–Wendroff (Richtmyer)

$$\text{Predictor: } u_{i+1/2}^{n+1/2} = \tfrac{1}{2}(u_i^n + u_{i+1}^n) - (\nu/2)(u_{i+1}^n - u_i^n)$$

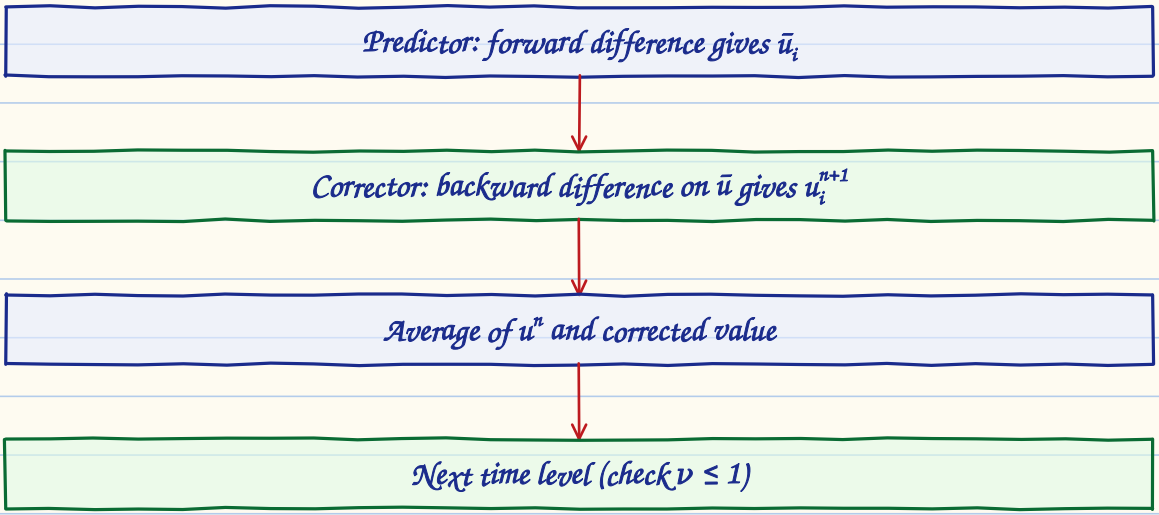
$$\text{Corrector: } u_i^{n+1} = u_i^n - \nu\,(u_{i+1/2}^{n+1/2} - u_{i-1/2}^{n+1/2})$$

Equivalent to Lax–Wendroff for linear problems but needs no Jacobian for nonlinear fluxes.

MacCormack predictor–corrector (very popular in CFD)

$$\text{Predictor: } \bar{u}_i = u_i^n - \nu\,(u_{i+1}^n - u_i^n) \quad \text{(forward difference)}$$

$$\text{Corrector: } u_i^{n+1} = \tfrac{1}{2}\,[u_i^n + \bar{u}_i - \nu\,(\bar{u}_i - \bar{u}_{i-1})] \quad \text{(backward difference)}$$



- Second order, stable for $\nu \le 1$. Alternating the direction of differences (F-B, B-F) removes bias; for linear problem it reduces to Lax–Wendroff.
- Widely used for Euler / Navier–Stokes (explicit, easy to code), with artificial viscosity added to damp oscillations at shocks.

5. Nonlinear Problems

Conservation-law form: $\partial u/\partial t + \partial f(u)/\partial x = 0$ with flux $f(u)$. Examples: Burgers $f = u^2/2$; Euler equations (vector u, vector f). Wave speed $a(u) = df/du$ depends on the solution.

- Characteristics: $dx/dt = a(u)$ — different values of u travel at different speeds -> wave steepening, characteristics converge -> shock (discontinuity) even from smooth initial data.
- Jump (Rankine–Hugoniot) condition for shock speed: $s = (f_R - f_L) / (u_R - u_L)$.
- Entropy condition selects the physical shock: $a(u_L) > s > a(u_R)$ (characteristics run into the shock).
- Use conservative difference forms $u_i^{n+1} = u_i^n - (\Delta t/\Delta x)(f_{i+1/2} - f_{i-1/2})$ so shocks have the correct speed

(Lax–Wendroff theorem). Non-conservative forms give wrong shock positions.

- Nonlinear Lax–Wendroff: $u_i^{n+1} = u_i^n - (\Delta t/2\Delta x)(f_{i+1} - f_{i-1}) + (\Delta t^2/2\Delta x^2)[A_{i+1/2}(f_{i+1} - f_i) - A_{i-1/2}(f_i - f_{i-1})]$, $A = \partial f/\partial u$.
- Stability (nonlinear): $\max|a|\,\Delta t/\Delta x \le 1$. Linearise about a local state and apply Von Neumann with frozen coefficients.
- Oscillations at shocks: add artificial viscosity $\varepsilon\,\Delta x^3\,|u_{xx}|\ldots$, use flux limiters / TVD schemes (higher order but monotone).

6. Second-order One-dimensional Wave Equation

Vibrating string, acoustic waves: $u_{tt} = c^2 u_{xx}$, $0 < x < L$, $u(x,0) = f(x)$, $u_t(x,0) = g(x)$, $u(0,t) = u(L,t) = 0$. Exact (d'Alembert): $u = \tfrac{1}{2}[f(x - ct) + f(x + ct)] + (1/2c)\int g\,ds$.

Explicit central-difference scheme

$$(u_i^{n+1} - 2u_i^n + u_i^{n-1})/\Delta t^2 = c^2\,(u_{i+1}^n - 2u_i^n + u_{i-1}^n)/\Delta x^2$$

$$\boxed{u_i^{n+1} = 2(1 - \nu^2)\,u_i^n + \nu^2(u_{i+1}^n + u_{i-1}^n) - u_i^{n-1}}$$

- Start-up: $u_i^0 = f_i$. The first step uses g: $u_i^1 = u_i^0 + \Delta t\,g_i + (\nu^2/2)(u_{i+1}^0 - 2u_i^0 + u_{i-1}^0)$.
- Von Neumann: $G + 1/G = 2(1 - 2\nu^2\sin^2(\theta/2))$ -> $|G| = 1$ if $|1 - 2\nu^2\sin^2(\theta/2)| \le 1$, i.e. $\nu = c\Delta t/\Delta x \le 1$. For $\nu = 1$ the scheme is exact on the grid.
- Second-order accurate, non-dissipative, conditionally stable (CFL).

Implicit scheme (Newmark / weighted-average)

$$(u^{n+1} - 2u^n + u^{n-1})/\Delta t^2 = c^2\,\delta_x^2/\Delta x^2\,[\beta\,u^{n+1} + (1 - 2\beta)\,u^n + \beta\,u^{n-1}]$$

- $\beta = 0$: explicit scheme above; $\beta \ge 1/4$: unconditionally stable (tridiagonal system each step); $\beta = 1/4$ is the average-acceleration (trapezoidal) method with no numerical damping.

Reduction to first-order system

Define $v = u_t$, $w = c\,u_x$: $v_t = c\,w_x$, $w_t = c\,v_x$ — a hyperbolic system $U_t + A\,U_x = 0$ with eigenvalues $\pm c$. Any first-order scheme (Lax–Wendroff, MacCormack) can be applied to U.

6A. Numerical Dissipation, Dispersion and the Modified Equation

Every scheme solves a slightly different PDE — the modified equation — obtained by expanding all terms in Taylor series about (x_i, t^n). For first-order upwind:

$$u_t + c\,u_x = (c\Delta x/2)(1 - \nu)\,u_{xx} - (c\Delta x^2/6)(1 - \nu)(1 - 2\nu)\,u_{xxx} + \ldots$$

- Even-derivative terms (u_{xx}, u_{xxxx}) -> dissipation: amplitude error, smearing of discontinuities, usually stabilising when the coefficient is positive.

- Odd-derivative terms (u_{xxx}) -> dispersion: different wavelengths travel at different speeds, giving wiggles (Gibbs-type oscillations), mostly behind a front for Lax–Wendroff.
- Exact amplification factor of the PDE: $|G_e| = 1$ and phase $-\nu\theta$. Compare numerical $|G|$ (dissipation error) and phase $\varphi = \arg G$ (dispersion error): relative phase error $\varphi/(-\nu\theta)$.

Scheme	Dominant error	Visible effect
Upwind / Lax	dissipation	profile smeared
Lax–Wendroff	dispersion	oscillations behind front
Leapfrog	dispersion	parasitic wave train
Godunov-type TVD	limited both	sharp, monotone

Godunov theorem and TVD limiters (brief)

- Godunov: a linear monotone scheme can be at most first-order accurate. To get sharp, oscillation-free shocks use non-linear schemes (TVD: total variation does not increase).
- Flux-limiter idea: blend upwind (monotone) and Lax–Wendroff (accurate): $\mathcal{F}_{i+1/2} = f^{up} + \psi(r)(f^{LW} - f^{up})$, r = ratio of consecutive gradients; e.g. minmod, van Leer, superbee limiters.

6B. Boundary Conditions for Hyperbolic Problems

- Specify conditions only where characteristics enter the domain. For $u_t + c\,u_x = 0$, $c > 0$: give u at the left (inflow) boundary; the right boundary needs no condition — use extrapolation / one-sided upwind scheme $u_N^{n+1} = u_N^n - \nu(u_N^n - u_{N-1}^n)$.
- Over-specification at outflow gives reflections and loss of well-posedness.
- Periodic problems: $u_0 \equiv u_N$ — convenient for testing schemes with Von Neumann analysis.
- For the wave equation $u_{tt} = c^2 u_{xx}$: fixed end $u = 0$, free end $u_x = 0$ (ghost node $u_{N+1} = u_{N-1}$); non-reflecting boundary $u_t + c\,u_x = 0$ at outflow.

6C. Systems of Hyperbolic Equations (1-D Euler / acoustics)

System: $U_t + A\,U_x = 0$. It is hyperbolic if the Jacobian A has real eigenvalues and a full set of eigenvectors. Diagonalise $A = R\Lambda R^{-1}$ and set $W = R^{-1}U$: $W_t + \Lambda\,W_x = 0$ — independent scalar waves (characteristic variables) with speeds λ_k.

- 1-D acoustics: $p_t + \rho c^2 u_x = 0$, $u_t + p_x/\rho = 0$: $\lambda = \pm c$. Apply the scheme to each characteristic variable, using upwind in the direction of each λ_k.
- 1-D Euler: $U = (\rho, \rho u, E)$, $\mathcal{F} = (\rho u, \rho u^2 + p, u(E + p))$; $\lambda = u, u + a, u - a$ (a = speed of sound).

CFL: $(|u| + a)\Delta t/\Delta x \le 1$.

- Applying a scalar scheme componentwise (e.g. Lax–Wendroff with matrix A) gives schemes for systems; stability requires $\rho(A)\Delta t/\Delta x \le 1$ where $\rho(A) = \max|\lambda|$.

6D. Two-dimensional Convection

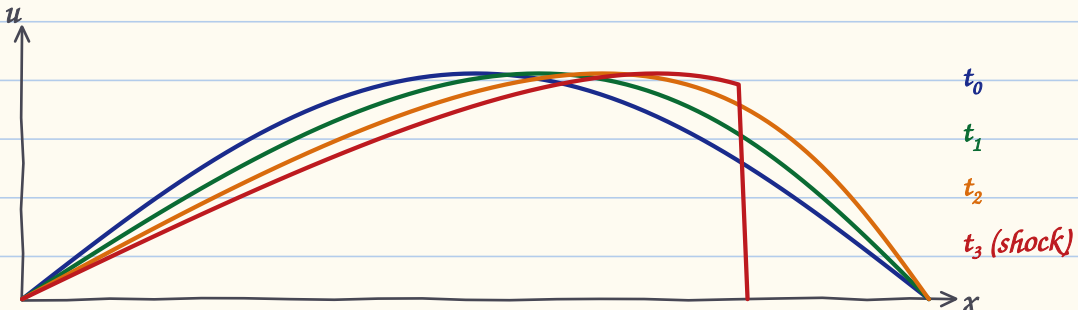
$u_t + a\,u_x + b\,u_y = 0$. Explicit upwind is stable for $|\nu_x| + |\nu_y| \le 1$ ($\nu_x = a\Delta t/\Delta x$, $\nu_y = b\Delta t/\Delta y$). Alternatives: fractional steps (solve x-sweep then y-sweep), 2-D Lax–Wendroff, MacCormack applied in each direction; implicit ADI for large time steps.

7. Burgers Equation

Simplest model that combines nonlinear convection and diffusion — a prototype for the Navier–Stokes momentum equation:

$$\partial u/\partial t + u\,\partial u/\partial x = \nu\,\partial^2 u/\partial x^2 \quad \text{(viscous Burgers)}$$

- $\nu = 0$: inviscid Burgers equation, hyperbolic, develops shocks ($f = u^2/2$, shock speed $s = (u_L + u_R)/2$).
- $\nu > 0$: parabolic; viscosity balances steepening -> smooth shock layer of thickness ~ $\nu/\Delta u$.
- Exact solution via Cole–Hopf: $u = -2\nu\,\varphi_x/\varphi$ transforms Burgers into the heat equation $\varphi_t = \nu\,\varphi_{xx}$.



steepening wave -> shock

Explicit scheme (FTCS)

$$u_i^{n+1} = u_i^n - (\Delta t/2\Delta x)\,u_i^n(u_{i+1}^n - u_{i-1}^n) + (\nu\Delta t/\Delta x^2)(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

- Stability (approximate, Von Neumann with frozen u): $2\nu\Delta t/\Delta x^2 \le 1$ and $(\Delta t/\nu)\,u^2 \le 2$ i.e. $\Delta t \le \min[\Delta x^2/2\nu\ ,\ 2\nu/u^2_{max}]$.
- Cell Reynolds number $Re_\Delta = u\Delta x/\nu \le 2$ required to avoid oscillations with central convection.
- Upwind convection alternative: $-(\Delta t/\Delta x)\,u_i^n(u_i^n - u_{i-1}^n)$ (for $u > 0$): more stable, more diffusive; conservative form $-(\Delta t/2\Delta x)(F_{i+1/2} - F_{i-1/2})$, $F = u^2/2$.

Implicit scheme

Fully implicit (BTCS) with linearised convection u_i^n (lagged coefficient):

$$-(r + C_i)\,u_{i-1}^{n+1} + (1 + 2r)\,u_i^{n+1} + (C_i - r)\,u_{i+1}^{n+1} = u_i^n$$

where $r = \nu\Delta t/\Delta x^2$ and $C_i = u_i^n\Delta t/2\Delta x$. Tridiagonal system -> TDMA each time step; unconditionally stable for the linearised form, large Δt allowed.

- Fully non-linear implicit ($u^{n+1}u_x^{n+1}$): solve by Newton iteration with a tridiagonal Jacobian, or by Picard (lagging) iteration within the time step.
- Crank–Nicolson version: average convective and diffusive terms at n and n+1; second order in time.

8. Runge–Kutta Methods (time integration)

Method of lines: discretise space first, $du/dt = \mathcal{F}(u)$ (a system of ODEs, e.g. $\mathcal{F} = -u\,u_x + \nu\,u_{xx}$ for Burgers), then integrate in time with a high-order explicit multistage scheme.

Second-order RK (Heun)

$$k_1 = \Delta t\,\mathcal{F}(u^n);\quad k_2 = \Delta t\,\mathcal{F}(u^n + k_1);\quad u^{n+1} = u^n + \tfrac{1}{2}(k_1 + k_2)$$

Classical fourth-order RK

$$u^{n+1} = u^n + (1/6)(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = \Delta t\mathcal{F}(u^n),\ k_2 = \Delta t\mathcal{F}(u^n + k_1/2),\ k_3 = \Delta t\mathcal{F}(u^n + k_2/2),\ k_4 = \Delta t\mathcal{F}(u^n + k_3)$$

Third-order TVD (Shu–Osher) — preferred for shocks

$$u^{(1)} = u^n + \Delta t\mathcal{F}(u^n);\quad u^{(2)} = \tfrac{3}{4}u^n + \tfrac{1}{4}[u^{(1)} + \Delta t\mathcal{F}(u^{(1)})];\quad u^{n+1} = \tfrac{1}{3}u^n + \tfrac{2}{3}[u^{(2)} + \Delta t\mathcal{F}(u^{(2)})]$$

- RK4 has a large stability region on the imaginary axis ($|\lambda\Delta t| \le 2.8$) so it works with central space differences for convection; RK2 is unstable for purely imaginary eigenvalues (central FTCS equivalent).
- Advantages: high-order in time, self-starting, easy to code, no linear solve. Disadvantage: several $\mathcal{F}$ evaluations per step; explicit time-step limit (CFL).

9. Worked Examples

Example 1 — Upwind scheme, step by step

Data: $c = 1$, $\Delta x = 0.1$, $\Delta t = 0.05$ -> $\nu = 0.5$. Initial: $u = 1$ for $x \le 0.2$, $u = 0$ beyond (step). Inflow $u = 1$.

Update $u_i^{n+1} = u_i^n - 0.5(u_i^n - u_{i-1}^n) = 0.5u_i^n + 0.5u_{i-1}^n$.

Nodes $x = 0.1, 0.2, 0.3, 0.4$: initial 1, 1, 0, 0. After step 1: 1, 1, 0.5, 0. After step 2: 1, 1, 0.75, 0.25.

The sharp step is smeared over several cells — numerical diffusion; exact solution would be a sharp step moving with speed 1.

Example 2 — Stability check by Von Neumann (upwind)

$G = 1 - \nu + \nu e^{-I\theta}$ -> $|G|^2 = (1 - \nu + \nu \cos\theta)^2 + \nu^2 \sin^2\theta = 1 - 2\nu(1 - \nu)(1 - \cos\theta)$.

$|G|^2 \le 1$ requires $\nu(1 - \nu) \ge 0$, i.e. $0 \le \nu \le 1$ √. For $\nu = 1.2$: $|G|^2 = 1 + 0.48(1 - \cos\theta) > 1$ — unstable.

Example 3 — Burgers explicit step

$\nu = 0.01$, $\Delta x = 0.1$, $\Delta t = 0.01$, $u_{i-1} = 1.0$, $u_i = 0.8$, $u_{i+1} = 0.5$.

Convective term: $-(0.01/0.2)(0.8)(0.5 - 1.0) = +0.02$. Diffusive term: $(0.01 \cdot 0.01/0.01)(0.5 - 1.6 + 1.0) = -0.001$.

$u_i^{n+1} = 0.8 + 0.02 - 0.001 = 0.819$. Check: $2\nu\Delta t/\Delta x^2 = 0.02 \le 1$ √, $Re_\Delta = 8 > 2$ — central convection may oscillate; reduce Δx or use upwind.

Stability regions of Runge–Kutta methods

For $du/dt = \lambda u$ with $z = \lambda\Delta t$, RK stability requires $|R(z)| \le 1$, where $R(z) = 1 + z + z^2/2$ (RK2), $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$ (RK4). Along the negative real axis: RK2 $z \ge -2$, RK4 $z \ge -2.78$; along the imaginary axis RK4: $|\text{Im } z| \le 2.83$ -> usable for convection with central differences when $\nu \le 2.8/(\max|\sin\theta|) \approx 2.8$.

10. Short Answer Questions

Q1. What is a hyperbolic PDE? -> $B^2 - 4AC > 0$; has real characteristics; signals travel at finite speed (wave equation, supersonic flow).

Q2. State the CFL condition. -> $\nu = c\Delta t/\Delta x \le 1$: numerical domain of dependence must contain the

physical one.

Q3. Why upwind differencing? -> information moves along characteristics, so the difference must use upstream points; yields stability and monotonicity at the cost of numerical diffusion.

Q4. What is numerical dissipation? -> amplitude damping from even-order truncation terms ($|G| < 1$); smears discontinuities.

Q5. What is numerical dispersion? -> phase error from odd-order truncation terms; different wavelengths move at different speeds.

Q6. Predictor–corrector advantage? -> second-order accuracy, explicit, no Jacobian matrices; MacCormack is the standard example.

Q7. Why conservative form for nonlinear laws? -> ensures correct shock speed through Rankine–Hugoniot jump conditions (Lax–Wendroff theorem).

Q8. Cell Reynolds number? -> $Re_\Delta = u\Delta x/\nu$; central scheme for Burgers/convection–diffusion stays oscillation-free only for $Re_\Delta \le 2$.

Q9. Leapfrog drawback? -> spurious computational mode and need for two starting levels; no damping to control it.

Q10. Why RK methods? -> high order in time, self-starting, easy; combined with method of lines for any spatial discretisation.

Comparison chart of schemes for $u_t + c\,u_x = 0$

Scheme	Explicit?	Levels	ν limit	Accuracy
Upwind	yes	2	1	1st
Lax	yes	2	1	1st
Lax–Wendroff	yes	2	1	2nd
Leapfrog	yes	3	1	2nd
MacCormack	yes	2 (2-stage)	1	2nd
BTCS	no	2	none	1st
Crank–Nicolson	no	2	none	2nd

Quick revision — Unit II

- Hyperbolic: finite speed, characteristics $dx/dt = c$; CFL $\nu = c\Delta t/\Delta x \le 1$.
- FTCS unstable; upwind (1st order, diffusive), Lax–Wendroff (2nd order, dispersive), leapfrog (non-dissipative), MacCormack (2-step).
- Implicit BTCS / Crank–Nicolson: unconditionally stable, tridiagonal.
- Wave equation $u_{tt} = c^2 u_{xx}$: explicit stable for $\nu \le 1$; Newmark $\beta \ge 1/4$ unconditionally stable.
- Burgers: nonlinear convection + diffusion; $Re_\Delta \le 2$ for central; RK2/RK4/TVD-RK3 for time integration.

Important Questions

1. Explain characteristics and the CFL condition. Derive the stability limits of upwind and Lax–Wendroff by Von Neumann analysis.
2. Why is FTCS unstable for $u_t + c\,u_x = 0$? How does the Lax scheme remove this?
3. Explain MacCormack and two-step Lax–Wendroff schemes.
4. Derive the explicit scheme for the wave equation $u_{tt} = c^2 u_{xx}$ and discuss its stability.
5. Write explicit and implicit schemes for Burgers equation; explain Runge–Kutta (RK4).

APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – III

Incompressible Viscous Flows and Compressible Flows

Syllabus (as given)

- Formulations of incompressible viscous flows by finite difference methods, pressure correction methods, vortex methods.
- Treatment of compressible flows: potential equation, Euler equations, Navier–Stokes system of equations, flow-field-dependent variation methods, boundary conditions, example problems.

1. Incompressible Viscous Flow — Governing Equations

For constant density ρ and viscosity μ (kinematic viscosity $\nu = \mu/\rho$):

$$\nabla \cdot \mathcal{V} = 0 \qquad \text{(continuity)}$$

$$\partial \mathcal{V}/\partial t + (\mathcal{V}\cdot\nabla)\mathcal{V} = -(1/\rho)\nabla p + \nu\nabla^2\mathcal{V} + f \qquad \text{(momentum)}$$

In 2-D Cartesian form:

$$u_x + v_y = 0$$

$$u_t + u\,u_x + v\,u_y = -p_x/\rho + \nu(u_{xx} + u_{yy})$$

$$v_t + u\,v_x + v\,v_y = -p_y/\rho + \nu(v_{xx} + v_{yy})$$

Difficulties of the incompressible equations

- There is no evolution equation for pressure — p acts as a Lagrange multiplier that enforces $\nabla\cdot\mathcal{V} = 0$, so continuity becomes a constraint, not a time-marching equation.
- Velocity–pressure coupling: u, v, p must be solved together; naive central differencing on a collocated grid leads to checkerboard pressure (odd–even decoupling).
- Non-linear convective terms ($u\,u_x$) and large Reynolds numbers $Re = UL/\nu$ need careful differencing (central vs upwind, cell Reynolds number).
- Two main strategies: (i) eliminate pressure — vorticity–stream function; (ii) keep primitive variables (u, v, p) — MAC, projection, SIMPLE (pressure correction).

2. Vorticity–Stream Function Formulation (2-D)

Define stream function ψ and vorticity ω:

$$u = \partial\psi/\partial y, \quad v = -\partial\psi/\partial x \qquad \text{(continuity automatically satisfied)}$$

$$\omega = \partial v/\partial x - \partial u/\partial y = -\nabla^2\psi$$

Take the curl of the momentum equations (pressure disappears):

$$\partial\omega/\partial t + u\,\partial\omega/\partial x + v\,\partial\omega/\partial y = \nu\,(\partial^2\omega/\partial x^2 + \partial^2\omega/\partial y^2) \qquad \text{(vorticity transport)}$$

$$\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = -\omega \qquad \text{(Poisson equation for } \psi\text{)}$$

Finite-difference discretisation

$$\omega_{ij}^{n+1} = \omega_{ij}^{n} - \Delta t[u\,(\omega_{i+1,j} - \omega_{i-1,j})/2\Delta x + v\,(\omega_{i,j+1} - \omega_{i,j-1})/2\Delta y] + \nu\Delta t[\delta_x^2\omega/\Delta x^2 + \delta_y^2\omega/\Delta y^2]$$

$$\psi_{i+1,j} + \psi_{i-1,j} + \psi_{i,j+1} + \psi_{i,j-1} - 4\psi_{i,j} = -h^2\,\omega_{ij} \qquad (\Delta x = \Delta y = h)$$

$$u_{ij} = (\psi_{i,j+1} - \psi_{i,j-1})/2\Delta y\,, \qquad v_{ij} = -(\psi_{i+1,j} - \psi_{i-1,j})/2\Delta x$$



Wall vorticity (Thom's formula)

At a stationary wall ψ = const (no flux) and $u = v = 0$. Expand ψ near the wall (wall at $j = 1$):

$$\omega_{wall} = -2(\psi_{wall+1} - \psi_{wall})/\Delta n^2 \qquad \text{(stationary wall)}$$

Moving lid with speed U: $\omega_w = -2(\psi_1 - \psi_w)/\Delta n^2 - 2U/\Delta n$

- Δn = distance between the wall and first interior node. Second-order accurate version uses three points.

Pros and cons of ψ–ω

- Pressure eliminated, continuity satisfied exactly, only 2 unknowns, good for 2-D cavity/channel flows.
- Wall vorticity BC is the weak point; not directly extendable to 3-D (vector potential needed); multiply-connected domains need extra conditions; pressure must be recovered separately by solving a Poisson equation.

Convective-term differencing in Navier–Stokes solvers

Scheme	Formula (u>0)	Remark
Central	$(u_{i+1} - u_{i-1})/2\Delta x$	2nd order; needs $Re_\Delta \le 2$
1st-order upwind	$(u_i - u_{i-1})/\Delta x$	stable, diffusive
2nd-order upwind	$(3u_i - 4u_{i-1} + u_{i-2})/2\Delta x$	less diffusive, wiggles
QUICK	$(3u_{i+1} + 3u_i - 7u_{i-1} + u_{i-2})/8\Delta x$	3rd order
Hybrid	central if $\|Pe\| < 2$, else upwind	robust

- Quasi-linear momentum equation: coefficient of u is itself u — treat by lagging (Picard) or Newton linearisation, with under-relaxation.
- Explicit stability limits for 2-D: $\Delta t \le \min(\Delta x^2/4\nu\ ,\ 4\nu/(u^2+v^2))$ for FTCS-type, and CFL $(|u|/\Delta x + |v|/\Delta y)\Delta t \le 1$ for upwind.

Worked example: wall vorticity

Thom formula — numbers

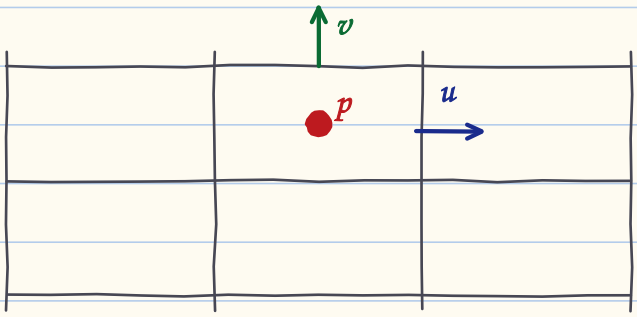
Stationary wall at $y = 0$, $\psi_{wall} = 0$, first interior node at $\Delta y = 0.1$ where $\psi_1 = 0.0004$.

$\omega_{wall} = -2(\psi_1 - \psi_{wall})/\Delta y^2 = -2(0.0004)/0.01 = -0.08$.

Moving lid $U = 1$ at the top, $\psi_{N-1} = -0.002$ (below the lid), $\Delta y = 0.1$: $\omega = -2(-0.002)/0.01 - 2(1)/0.1 = +0.4 - 20 = -19.6$.

3. Primitive-variable Formulations

3.1 Staggered grid (Harlow–Welch)



p at cell centre

u at east/west faces

v at north/south faces

→ avoids checkerboard p

- Pressure at cell centres, u at vertical faces, v at horizontal faces -> pressure difference drives the velocity located between the two pressure nodes; discrete continuity involves compact differences

$(u_{i+1/2} - u_{i-1/2})/\Delta x + (v_{j+1/2} - v_{j-1/2})/\Delta y$.

- Eliminates checkerboard oscillation; widely used in finite-difference and finite-volume codes (alternative: collocated grid + Rhie–Chow interpolation).

3.2 Pressure Poisson equation

Take divergence of momentum equation and use $\nabla \cdot \mathcal{V} = 0$:

$$\nabla^2 p = -\rho \left[(\partial u/\partial x)^2 + 2\,(\partial u/\partial y)(\partial v/\partial x) + (\partial v/\partial y)^2 \right] \quad (+\,\partial D/\partial t \text{ term in MAC})$$

- With Neumann conditions on walls $\partial p/\partial n = \mu\, \nabla^2 \mathcal{V} \cdot n$. Solved by SOR / multigrid at every step.

3.3 MAC (Marker-And-Cell) method

1. Given $\mathcal{V}^n$ (with $\nabla \cdot \mathcal{V}^n = 0$), compute pressure p^n from the Poisson equation.
2. Advance momentum explicitly: $\mathcal{V}^{n+1} = \mathcal{V}^n + \Delta t[-(\mathcal{V} \cdot \nabla)\mathcal{V} - \nabla p/\rho + \nu \nabla^2 \mathcal{V}]$.
3. Marker particles are moved with the new velocity to track the free surface.

3.4 Fractional-step (Chorin projection) method

Step 1 Intermediate velocity ignoring pressure: $\mathcal{V}^* = \mathcal{V}^n + \Delta t[-(\mathcal{V}^n \cdot \nabla)\mathcal{V}^n + \nu \nabla^2 \mathcal{V}^n]$

Step 2 Pressure Poisson: $\nabla^2 p^{n+1} = (\rho/\Delta t) \nabla \cdot \mathcal{V}^*$

Step 3 Projection (correction): $\mathcal{V}^{n+1} = \mathcal{V}^* - (\Delta t/\rho) \nabla p^{n+1}$ -> $\nabla \cdot \mathcal{V}^{n+1} = 0$

The pressure projects $\mathcal{V}^*$ onto the space of divergence-free fields. Time step limits (explicit): $\Delta t \le \Delta x^2/4\nu$ and $\Delta t \le 4\nu/u^2$ (also CFL).

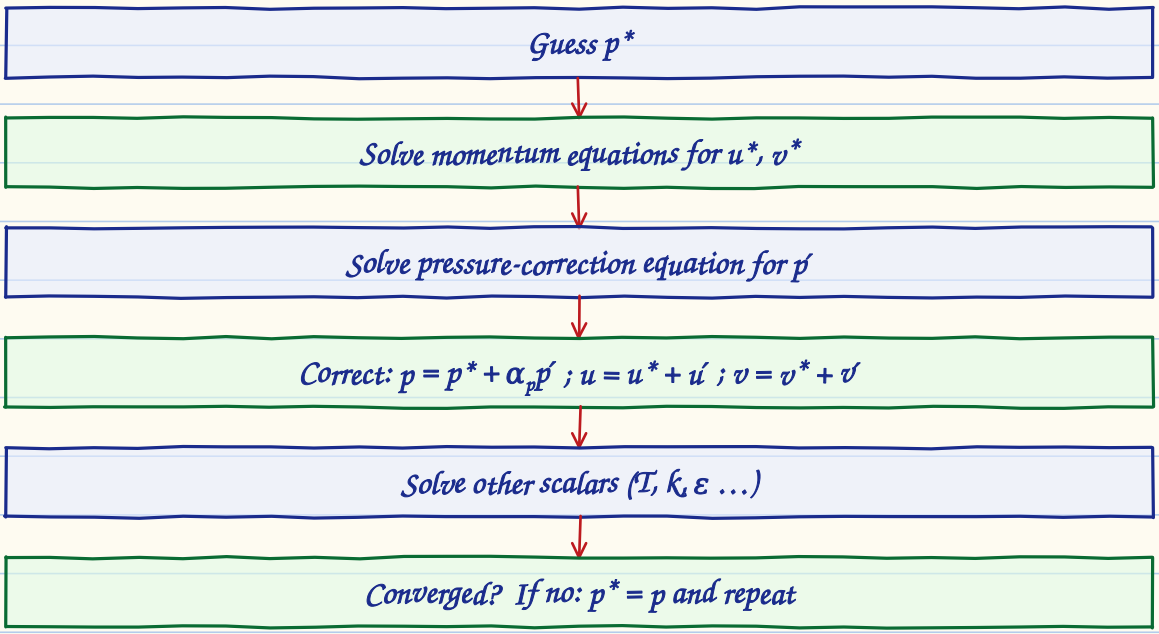
4. Pressure Correction Methods — SIMPLE family

Used in finite-volume/finite-difference codes for steady (and unsteady) flow. Idea: guess pressure, solve momentum, then correct pressure and velocity so that continuity is satisfied.

SIMPLE = Semi-Implicit Method for Pressure-Linked Equations (Patankar & Spalding, 1972)

$$p = p^* + p' , \quad u = u^* + u' , \quad v = v^* + v'$$

- Starred values: from the momentum equations with guessed pressure p^*. Primed values: corrections.
- Velocity correction: $u' = -d_e (p'_E - p'_P)$, $d_e = A_e/a_e$ (neglecting neighbour corrections — the "simplification" in SIMPLE).
- Substitute into continuity -> pressure-correction equation: $a_P p'_P = a_E p'_E + a_W p'_W + a_N p'_N + a_S p'_S + b$, b = mass imbalance of u^*, v^*.



- Under-relaxation is essential: $\alpha_p \approx 0.3$ for pressure and $\alpha_u \approx 0.7$ for velocity (steady problems); $\alpha_p + \alpha_u \approx 1$ for optimum.
- SIMPLER: solves a pressure equation (not correction) for p, uses p′ only to correct velocities; better convergence.
- SIMPLEC: approximate d by $\mathcal{A}/(a_P - \Sigma a_{nb})$ — keeps neighbour corrections; α_p can be 1; faster.
- PISO (Pressure-Implicit with Splitting of Operators): one predictor + two or more corrector steps per time step; best for transient flows, no under-relaxation needed.

Method	Correctors	Under-relax	Best use
SIMPLE	1	needed	steady flow
SIMPLER	1 + p-eq	needed	steady, better conv.
SIMPLEC	1 (consistent)	light	steady
PISO	2 or more	not needed	transient

Discretised momentum equation in SIMPLE (finite-volume form)

$$a_e u_e = \Sigma\, a_{nb} u_{nb} + b + (p_P - p_E)\mathcal{A}_e$$

a_{nb} from convection–diffusion (upwind / hybrid / power law); $(p_P - p_E)\mathcal{A}_e$ = pressure force across the east face of the u control volume (u-cell is staggered). Pressure-correction coefficients:

$$a_E = \rho_e d_e \mathcal{A}_e, \quad a_W = \rho_w d_w \mathcal{A}_w, \quad a_P = a_E + a_W + a_N + a_S, \quad b = (\rho u^*)_w \mathcal{A}_w - (\rho u^*)_e \mathcal{A}_e + (\rho v^*)_s \mathcal{A}_s - (\rho v^*)_n \mathcal{A}_n$$

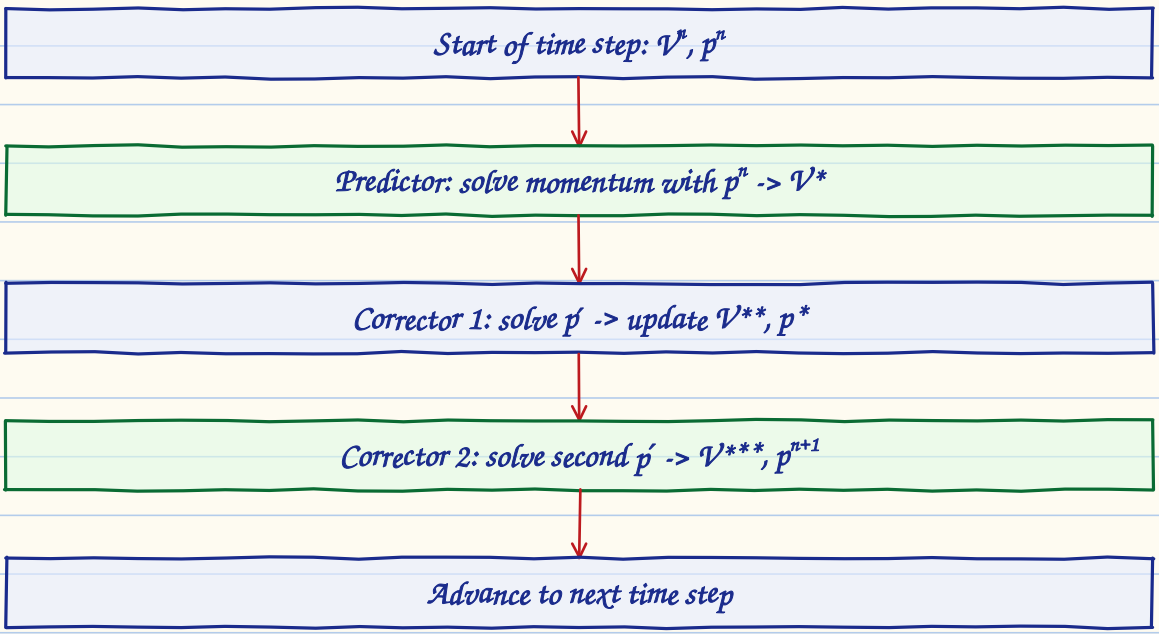
- b is the mass source: when $b \to 0$ everywhere, continuity is satisfied and the solution is converged.
- Convergence monitors: residuals of u, v, mass imbalance ($\Sigma |b|$), and key quantity such as drag. Typical criterion: normalised residual < 10^{-4} to 10^{-6}.

- Boundary conditions for p' : at an inlet/wall where velocity is specified, $u' = 0$ so no p' equation is needed (Neumann).

Example: 1-D pressure correction

Mass imbalance gives p'

Cell with $u^*_w = 1.0$, $u^*_e = 1.2$ ($\rho = 1$, $A = 1$). $b = 1.0 - 1.2 = -0.2$ (more leaves than enters). With $d_e = d_w = 0.5$: $a_E = a_W = 0.5$, $a_P = 1.0$ and $p'_E = p'_W = 0$: $p'_P = -0.2$. Negative p'_P reduces the outflow and increases inflow -> mass imbalance is removed.



5. Vortex Methods

Lagrangian, grid-free approach: the flow is represented by a collection of discrete vortices (vortex blobs) that move with the local velocity. Pressure does not appear.

- Vorticity transport equation (2-D): $D\omega/Dt = \nu\nabla^2\omega$. Inviscid part: vorticity is carried by the flow; viscous part: diffusion of vorticity.
- Discretise: $\omega(x) \approx \Sigma_j \Gamma_j \zeta_\sigma(x - x_j)$ with circulation Γ_j and smoothing core ζ_σ.
- Velocity from the Biot–Savart law:

$$V(x) = \Sigma_{j=1}^{N} \Gamma_j K_\sigma(x - x_j), \quad K(x) = (1/2\pi)(-y, x)/|x|^2$$

- Regularised kernel (Chorin blob): $K_\sigma = K\cdot(|x|^2/(|x|^2 + \sigma^2))$ to avoid singularity when vortices come close.
- Convection step: $dx_i/dt = V(x_i)$, integrated by RK2 / Euler. Cost $O(N^2)$ — reduced by fast multipole / tree codes.

Viscous effects in vortex methods

- **Random-walk method** (Chorin): each blob is given a random displacement $\Delta x, \sim \mathcal{N}(0, 2\nu\Delta t)$ each step — stochastic simulation of diffusion.
- Core-spreading and particle-strength-exchange (PSE) methods give deterministic diffusion.
- Boundary layers: new vortex sheets are created at a wall to satisfy no-slip.
- Vortex-in-cell (VIC): vorticity assigned to a grid, Poisson equation $\nabla^2\psi = -\omega$ solved on the grid (FFT) and velocities interpolated back — $O(N \log N)$.

Features

- Advantages: no numerical diffusion, no grid for the flow region (good for external flows, wakes, high Re), automatic far-field BC.
- Limitations: expensive N-body sum, boundary treatment, statistical noise in random walk, compressibility not included.

Algorithm of a vortex-blob code

Initialise blobs (positions x_j, circulations Γ_j)

↓

Compute velocity at each blob by Biot–Savart sum

↓

Move blobs: $x_j \leftarrow x_j + V\Delta t$ (RK2)

↓

Add random walk $\xi \sim \mathcal{N}(0, 2\nu\Delta t)$ for viscosity

↓

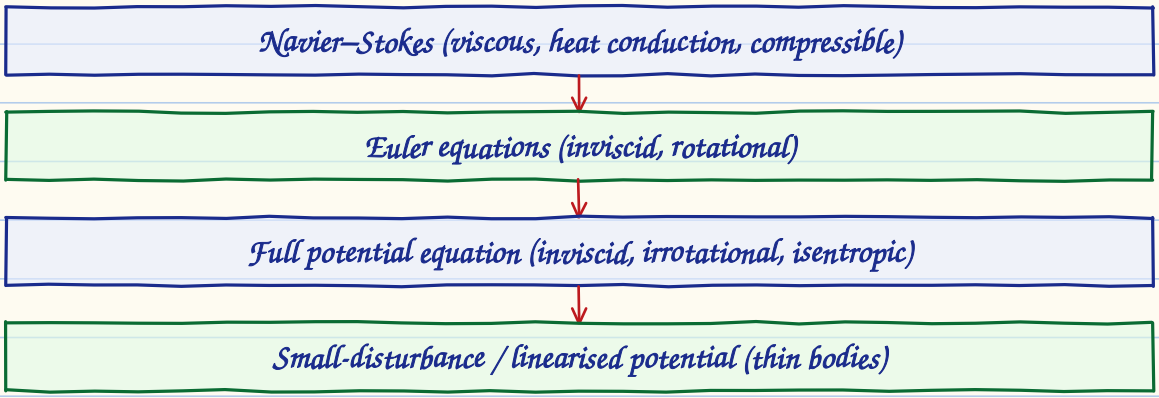
Create new vortices at the wall (no-slip) and merge/remove blobs

Example: two equal point vortices

Two identical vortices of circulation Γ at distance d. Each induces speed $V = \Gamma/(2\pi d)$ on the other, perpendicular to the line joining them.

They rotate about their midpoint with angular velocity $\Omega = V/(d/2) = \Gamma/(\pi d^2)$. Opposite signs (vortex pair) translate together with speed $\Gamma/(2\pi d)$.

6. Compressible Flows — Hierarchy of Models



6.1 Potential equation

For irrotational flow $\mathbf{V} = \nabla\varphi$, isentropic, steady, the continuity equation becomes the full potential equation:

$$(a^2 - \varphi_x^2)\varphi_{xx} + (a^2 - \varphi_y^2)\varphi_{yy} - 2\varphi_x\varphi_y\varphi_{xy} = 0\,, \quad a^2 = a_0^2 - ((\gamma-1)/2)(\varphi_x^2 + \varphi_y^2)$$

- Single non-linear second-order PDE. Type: elliptic for subsonic ($M < 1$), hyperbolic for supersonic ($M > 1$) — a mixed type problem for transonic flow.
- Linearised small-disturbance form $(1 - M_\infty^2)\varphi_{xx} + \varphi_{yy} = 0$ (Prandtl–Glauert).
- Type-dependent differencing (Murman–Cole): central differences in subsonic points; upwind (retarded density) in supersonic points; solved by line SOR; conservative form needed to capture shocks. Isentropic assumption overestimates shock strength.

6.2 Euler equations

Inviscid, conservative form (2-D):

$$\partial U/\partial t + \partial F/\partial x + \partial G/\partial y = 0$$

$$U = (\rho, \rho u, \rho v, E)^T\,, \quad F = (\rho u, \rho u^2 + p, \rho uv, u(E + p))^T\,, \quad G = (\rho v, \rho uv, \rho v^2 + p, v(E + p))^T$$

$$E = \rho[e + (u^2 + v^2)/2]\,, \quad p = (\gamma - 1)[E - \tfrac{1}{2}\rho(u^2 + v^2)]$$

- Hyperbolic in time with eigenvalues of the Jacobian $A = \partial F/\partial U$: $u,\ u,\ u + a,\ u - a$ ($a = \sqrt{(\gamma p/\rho)}$).
- Solved by explicit MacCormack, Lax–Wendroff, Jameson RK schemes with artificial dissipation, or upwind / Godunov / Roe schemes (see Unit II, IV). CFL: $\Delta t \le CFL\cdot\Delta x/(|u| + a)$.

6.3 Navier–Stokes system (compressible)

$$\partial U/\partial t + \partial F/\partial x + \partial G/\partial y = \partial F_v/\partial x + \partial G_v/\partial y$$

$$F_v = (0, \tau_{xx}, \tau_{xy}, u\tau_{xx} + v\tau_{xy} - q_x)^T\,, \quad G_v = (0, \tau_{xy}, \tau_{yy}, u\tau_{xy} + v\tau_{yy} - q_y)^T$$

$$\tau_{xx} = \mu(4u_x - 2v_y)/3\,, \quad \tau_{yy} = \mu(4v_y - 2u_x)/3\,, \quad \tau_{xy} = \mu(u_y + v_x)\,, \quad q = -k\nabla T$$

- Closed with perfect-gas law and Sutherland viscosity $\mu(T)$. Convective part treated as hyperbolic

(MacCormack / upwind), viscous part by central differences.

- Explicit MacCormack for N–S: predictor-corrector with forward/backward differences on convective terms, alternating on viscous derivatives (cross derivatives central). Time step: $\Delta t \le \min(\Delta t_{CFL}, \Delta t_{visc})$, $\Delta t_{visc} \sim \Delta x^2 \rho / 2\mu$.

7. Flow-Field-Dependent Variation (FDV) Methods

Developed by T. J. Chung. Aim: a single finite element / finite volume algorithm valid for the whole range of flows: incompressible → compressible, laminar → turbulent, subsonic → hypersonic.

1. Expand the unknown in a Taylor series in time including second-order term: $U^{n+1} = U^n + \Delta t\, U_t^n + (\Delta t^2/2)\, U_{tt}^n + \dots$
2. Replace the time derivatives by spatial derivatives using the governing equations ($U_t = -\nabla \cdot \mathcal{F} + \dots$, $U_{tt} = \dots$).
3. Introduce flow-field-dependent variation parameters s_1, s_2 (convection) and s_3 (diffusion) that are computed from the local gradients of the flow variables between two iterations: $\Delta U/\Delta t$ and $\Delta \mathcal{F}/\Delta U$ — i.e. from the change of the flow field itself.
4. The resulting terms act as automatic, solution-dependent artificial viscosity — large at shocks and steep gradients, small in smooth regions — so no ad-hoc dissipation coefficients are required.
5. Equations are solved on a finite element (or finite volume) mesh with the Galerkin method and an implicit/explicit time integration (Newton / GMRES).

- Result: shock capturing without user-tuned smoothing, a unified treatment of pressure–velocity coupling (compressibility enters through the equation of state), efficient on unstructured meshes.

Shock relations (normal shock) — useful for checks

$$p_2/p_1 = 1 + (2\gamma/(\gamma+1))(M_1^2 - 1), \quad \rho_2/\rho_1 = (\gamma+1)M_1^2/((\gamma-1)M_1^2 + 2)$$

$$M_2^2 = (1 + (\gamma-1)M_1^2/2) / (\gamma M_1^2 - (\gamma-1)/2)$$

Numerical solutions are validated against these Rankine–Hugoniot values; conservative schemes reproduce them, non-conservative schemes do not.

Artificial viscosity for central/Lax–Wendroff-type Euler solvers

- Second-difference term (shock capturing): $\varepsilon^{(2)}(U_{i+1} - 2U_i + U_{i-1})$, $\varepsilon^{(2)} \propto |p_{i+1} - 2p_i + p_{i-1}|/(p_{i+1} + 2p_i + p_{i-1})$ — large only across shocks.
- Fourth-difference term (background damping in smooth regions): $-\varepsilon^{(4)}(U_{i+2} - 4U_{i+1} + 6U_i - 4U_{i-1} + U_{i-2})$. Jameson–Schmidt–Turkel (JST) scheme uses both with RK time stepping.

MacCormack algorithm for 2-D Euler / N–S

Predictor: $\bar{U} = U^n - (\Delta t/\Delta x)\Delta F^n_{forward} - (\Delta t/\Delta y)\Delta G^n_{forward}$

↓

Evaluate $F, \bar{G}$ from $\bar{U}$ (new fluxes)

↓

Corrector: $U^{n+1} = \frac{1}{2}[U^n + \bar{U} - (\Delta t/\Delta x)\Delta F_{backward} - (\Delta t/\Delta y)\Delta \bar{G}_{backward}]$

↓

Add artificial viscosity; apply BCs; set Δt from CFL

Short answer questions

Q1. Why is the pressure difficult in incompressible flow? -> no evolution equation; it only enforces $\nabla \cdot V = 0$.

Q2. Why a staggered grid? -> avoids checkerboard pressure fields by coupling p and u directly.

Q3. Why under-relaxation in SIMPLE? -> p' overcorrects (neglected neighbour terms) leading to divergence.

Q4. Advantage of vortex methods? -> grid-free, no numerical diffusion, pressure eliminated.

Q5. Type of the potential equation? -> elliptic if $M < 1$, hyperbolic if $M > 1$ (mixed in transonic flow).

Q6. Why conservation form for Euler equations? -> correct shock speeds/strengths (Rankine–Hugoniot).

8. Boundary Conditions for Flow Problems

Boundary	Condition	Remarks
Solid wall (viscous)	$u = v = 0$; $T = T_w$ or $\partial T/\partial n = 0$	no-slip; adiabatic / isothermal
Solid wall (inviscid)	$V \cdot n = 0$	slip; tangency
Symmetry plane	$V \cdot n = 0$, $\partial \varphi/\partial n = 0$	half-domain only
Inflow (subsonic)	specify ρ, u, v (or total p, T)	one variable from inside
Outflow (subsonic)	specify static p	extrapolate rest
Supersonic inflow	all variables fixed	characteristics all enter
Supersonic outflow	extrapolate all	no info from outside
Far-field	characteristic / freestream	non-reflecting

- Number of conditions = number of characteristics entering the domain (from the sign of eigenvalues u, $u \pm a$).
- Incompressible pressure: Neumann at walls $\partial p/\partial n = \ldots$; at least one Dirichlet value (or reference) needed since p is defined up to a constant.
- Numerical treatment: ghost cells, one-sided differences, extrapolation $\varphi_N = 2\varphi_{N-1} - \varphi_{N-2}$; wall pressure from normal momentum $\partial p/\partial n = 0$ (boundary layer).

9. Example Problems

Example 1 — Lid-driven cavity (incompressible, ψ–ω)

Square cavity, lid speed $U = 1$ along $y = 1$, $Re = UL/\nu = 100$. Boundary: $\psi = 0$ on all walls; ω on walls from Thom's formula (lid: $-2\psi_1/\Delta y^2 - 2U/\Delta y$).

Grid 41×41, $\Delta t = 0.005$ (explicit, $\Delta t \le \Delta x^2/4\nu$), SOR with $\omega_s \approx 1.8$ for Poisson. Result: primary vortex centre near (0.62, 0.74), small corner vortices at the bottom corners (Ghia et al. benchmark).

Example 2 — Shock tube (Sod problem), 1-D Euler

Initial: left ($x < 0.5$): $\rho = 1$, $u = 0$, $p = 1$; right: $\rho = 0.125$, $u = 0$, $p = 0.1$; $\gamma = 1.4$.

Solution contains a left-moving rarefaction fan, a contact discontinuity and a right-moving shock. MacCormack with artificial viscosity or TVD scheme captures all waves; Δt from CFL ≤ 0.8.

Example 3 — Supersonic flow past a wedge

$M_\infty = 2$, wedge half-angle 10°: oblique shock (θ-β-M relation) gives $\beta \approx 39.3°$, downstream $M \approx 1.64$. Euler solver marching in x (hyperbolic in supersonic flow); wall: $V \cdot n = 0$; inflow all fixed; outflow extrapolated.

Quick revision — Unit III

- Incompressible: ψ–ω (no pressure, wall ω from Thom), MAC/projection (Poisson for p), SIMPLE ($p = p^* + p'$, under-relaxation).
- Vortex methods: Biot–Savart law, blobs, random walk for viscosity, grid-free.
- Compressible: potential (elliptic/hyperbolic mixed) → Euler (hyperbolic) → Navier–Stokes; eigenvalues u, $u \pm a$.
- FDV: Taylor-series in time with flow-dependent parameters s_1, s_2, s_3 = built-in shock-capturing dissipation.
- BCs: number of conditions = number of characteristics entering.

Important Questions

1. Derive the vorticity–stream function formulation and explain how wall vorticity is obtained.
2. Explain the SIMPLE algorithm with a flow chart; compare with SIMPLER, SIMPLEC and PISO.
3. What is the staggered grid and why is it used? Explain the projection method.
4. Explain vortex methods with Biot–Savart law and the treatment of viscosity.
5. Write the Euler and Navier–Stokes equations in conservation form; explain FDV methods and boundary conditions.

APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – IV

Finite Volume Method

Syllabus (as given)

- Finite volume method via finite difference method.
- Formulations for two- and three-dimensional problems.

1. Basic Idea of the Finite Volume Method (FVM)

FVM = integrate the governing conservation equation over each small control volume (CV) of the grid, then convert volume integrals of divergence terms into surface integrals (fluxes) using Gauss' divergence theorem. The discrete equations express exact conservation of mass, momentum, energy over every CV.

$$\int_{CV} \nabla\cdot(\rho V\varphi)\, dV = \int_{A} (\rho V\varphi)\cdot n\, dA = \Sigma_{faces}\, F_f \varphi_f \quad \text{(Gauss theorem)}$$

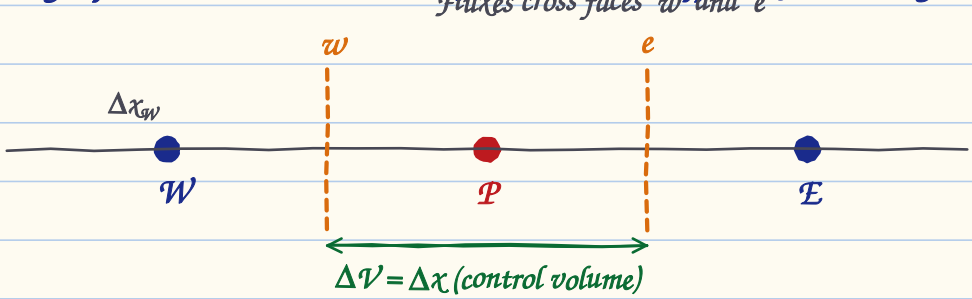
General transport equation

$$\partial(\rho\varphi)/\partial t + \nabla\cdot(\rho V\varphi) = \nabla\cdot(\Gamma\nabla\varphi) + S_\varphi$$

Integrate over CV and over time Δt:

$$\int_{\Delta t}[\partial/\partial t \int_{CV} \rho\varphi\, dV + \int_{A} n\cdot(\rho V\varphi)\, dA\,]\, dt = \int_{\Delta t}[\int_{A} n\cdot(\Gamma\nabla\varphi)\, dA + \int_{CV} S_\varphi\, dV\,]\, dt$$

i.e. (rate of change of φ in CV) + (net convective flux out) = (net diffusive flux in) + (generation).



Steps of FVM

1. Grid generation: divide domain into non-overlapping CVs; nodes P at CV centre; faces w, e (n, s, t, b).
2. Integrate the conservation equations over each CV.
3. Discretisation: approximate face values and gradients by interpolation (profile assumptions) -> algebraic equation per CV.
4. Assemble and solve algebraic equations (TDMA, Gauss–Seidel, multigrid); iterate for

non-linearities.

Why FVM?

- Conservative by construction (flux leaving one CV = flux entering neighbour).
- Works on structured, unstructured and non-orthogonal meshes; handles complex geometry.
- Physical interpretation of every term; basis of FLUENT, CFX, OpenFOAM, STAR-CD.

2. FVM for 1-D Steady Diffusion

Equation: $d/dx(\Gamma\, d\varphi/dx) + S = 0$ (e.g. heat conduction with $\Gamma = k$). Integrate over CV ($\Delta x = x_e - x_w$):

$$(\Gamma A\, d\varphi/dx)_e - (\Gamma A\, d\varphi/dx)_w + S\,\Delta V = 0$$

Face gradient by central (linear) profile between nodes: $(d\varphi/dx)_e = (\varphi_E - \varphi_P)/\delta x_{PE}$, $(d\varphi/dx)_w = (\varphi_P - \varphi_W)/\delta x_{WP}$. Source linearised: $S\Delta V = S_u + S_P\varphi_P$ ($S_P \le 0$).

$$a_P\varphi_P = a_E\varphi_E + a_W\varphi_W + b$$

Coefficient	Value	Meaning
a_W	$\Gamma_w A_w/\delta x_{WP}$	west conductance
a_E	$\Gamma_e A_e/\delta x_{PE}$	east conductance
a_P	$a_W + a_E - S_P$	centre coefficient
b	S_u	constant source

- Face conductivity for non-uniform Γ: harmonic mean $\Gamma_e = 2\Gamma_P\Gamma_E/(\Gamma_P + \Gamma_E)$ (keeps flux continuity across material interface); for uniform grid, linear interpolation $\Gamma_e = \tfrac{1}{2}(\Gamma_P + \Gamma_E)$.

Boundary nodes

- Dirichlet φ_B given at boundary: boundary-face gradient over half cell: $(\varphi_P - \varphi_B)/(\delta x/2)$ -> extra coefficient $a_B = 2\Gamma A/\delta x$: $S_u = a_B\varphi_B$, $S_P = -a_B$.
- Neumann (given flux q): add $q\,A$ to source term, set boundary a-coefficient to zero. Insulated: flux = 0.

Worked example — 1-D conduction in a rod

Rod $L = 0.5$ m, $A = 10^{-2}$ m², $k = 1000$ W/mK, $T_A = 100°C$, $T_B = 500°C$, no source. Use 5 CVs, $\Delta x = 0.1$ m.

Interior: $a_W = a_E = kA/\Delta x = 100$. Node 1: $a_W = 0$, $a_E = 100$, $S_P = -200$, $S_u = 200T_A$ -> $a_P = 300$. Node 5: $S_u = 200T_B$, $a_P = 300$.

Equations: $300T_1 = 100T_2 + 200(100)$; $200T_2 = 100T_1 + 100T_3$; … ; $300T_5 = 100T_4 + 200(500)$.

Solution (TDMA): $T = 140, 220, 300, 380, 460$ °C = exact line $T = 100 + 800x$ at $x = 0.05, 0.15, \ldots$ √

Source-term linearisation

Non-linear source $S(\varphi)$ is linearised about the current value φ^*: $S = S_C + S_P\varphi_P$ with $S_P \le 0$ so that a_P stays positive (diagonal dominance).

$$S(\varphi) \approx S^* + (dS/d\varphi)^*(\varphi_P - \varphi^*) \quad \rightarrow \quad S_P = (dS/d\varphi)^* \le 0\,, \quad S_C = S^* - (dS/d\varphi)^*\varphi^*$$

Case	Treatment
$S = 4 - 5\varphi$	$S_C = 4$, $S_P = -5$ (already linear, negative slope)
$S = 3 + 0.2\varphi$ (positive slope)	put all in $S_C = 3 + 0.2\varphi^*$ (explicit, lagged), $S_P = 0$
$S = x\varphi^2$ … (non-linear)	Newton: $S_P = -2x\varphi^*$, $S_C = x\varphi^{*2}$

3. FVM for 1-D Convection–Diffusion

Steady: $d(\rho u\varphi)/dx = d/dx(\Gamma\, d\varphi/dx)$, with continuity $d(\rho u)/dx = 0$. Define convective strength $F = \rho u$ and diffusion conductance $D = \Gamma/\delta x$ at faces:

$$F_e\varphi_e - F_w\varphi_w = D_e(\varphi_E - \varphi_P) - D_w(\varphi_P - \varphi_W)$$

Peclet number $Pe = F/D = \rho u \delta x/\Gamma$ (ratio of convection to diffusion). The remaining task: express φ_e, φ_w in terms of nodal values.

(a) Central differencing

$$\varphi_e = (\varphi_P + \varphi_E)/2\,, \quad \varphi_w = (\varphi_W + \varphi_P)/2$$

$$a_E = D_e - F_e/2\,, \quad a_W = D_w + F_w/2\,, \quad a_P = a_E + a_W + (F_e - F_w)$$

- Second-order accurate, but $a_E < 0$ when $Pe > 2$ -> unbounded, oscillating (wiggles) solution. Valid only for $|Pe| < 2$.

(b) Upwind differencing

$$\varphi_e = \varphi_P \ (\mathcal{F} > 0), \quad \varphi_e = \varphi_E \ (\mathcal{F} < 0)$$

$$a_W = \mathcal{D}_w + \max(\mathcal{F}_w, 0), \quad a_E = \mathcal{D}_e + \max(-\mathcal{F}_e, 0)$$

- Always positive coefficients -> bounded and stable for all Pe; only first-order accurate; false (numerical) diffusion $\Gamma_{false} \approx \rho u \Delta x/2$.

(c) Hybrid scheme (Spalding)

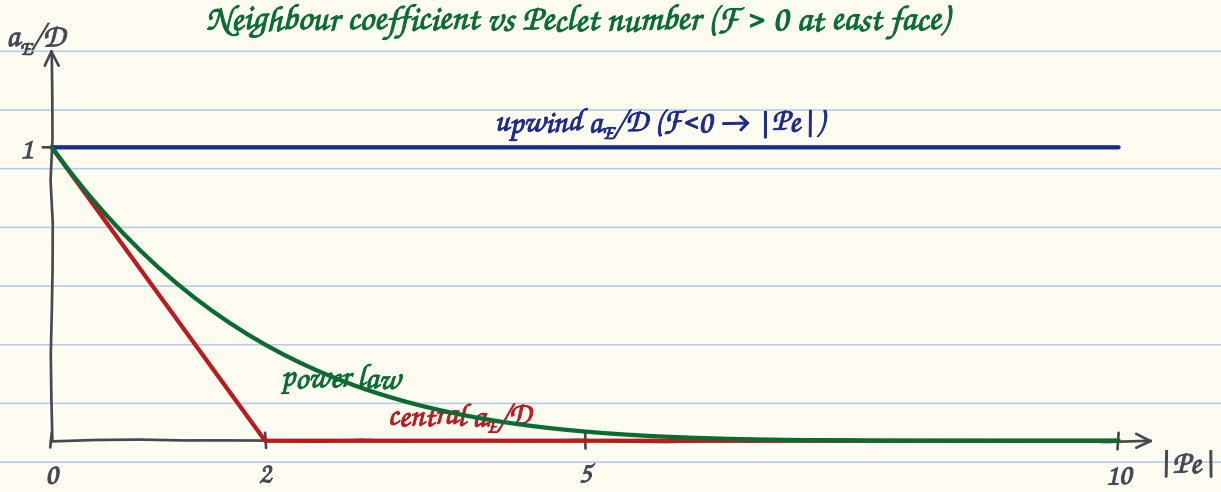
$$a_E = \max(-\mathcal{F}_e, \mathcal{D}_e - \mathcal{F}_e/2, 0), \quad a_W = \max(\mathcal{F}_w, \mathcal{D}_w + \mathcal{F}_w/2, 0)$$

- Central differencing for $|Pe| < 2$; upwind with diffusion set to zero for $|Pe| \geq 2$.

(d) Power-law and QUICK

- Power-law (Patankar): $a_E = \mathcal{D}_e \max(0, (1 - 0.1|Pe_e|)^5) + \max(-\mathcal{F}_e, 0)$ — close to the exact 1-D solution.
- QUICK (Leonard): quadratic upstream interpolation $\varphi_e = (6/8)\varphi_P + (3/8)\varphi_E - (1/8)\varphi_W$ $(\mathcal{F} > 0)$; third-order accurate, may be unbounded.

Neighbour coefficient vs Peclet number ($\mathcal{F} > 0$ at east face)



Three basic rules (Patankar)

- Conservativeness: flux through a common face is the same for both neighbouring CVs.
- Boundedness: all coefficients positive ($a_P = \Sigma a_{nb}$) — no overshoot.
- Transportiveness: influence of neighbours must depend on Pe (upstream influence grows with Pe).

Scheme	Order	Bounded	Remark
Central	2	$Pe < 2$ only	wiggles
Upwind	1	always	false diffusion
Hybrid	1	always	robust, simple
Power law	~1	always	accurate 1-D
QUICK	3	not always	needs limiter

Exact solution and worked example

Exact solution of the 1-D steady convection–diffusion problem with $\varphi = \varphi_0$ at $x = 0$ and $\varphi = \varphi_L$ at $x = L$:

$$(\varphi - \varphi_0)/(\varphi_L - \varphi_0) = (\exp(Pe \cdot x/L) - 1)/(\exp(Pe) - 1), \qquad Pe = \rho u L/\Gamma$$

Example: $\rho = 1, \Gamma = 0.1, L = 1$, 5 equal cells

$\delta x = 0.2$: $D = \Gamma/\delta x = 0.5$. $u = 0.1 \rightarrow F = 0.1$, $Pe_{cell} = F/D = 0.2 < 2$: central differencing is acceptable.

Interior coefficients (central): $a_W = D + F/2 = 0.55$, $a_E = D - F/2 = 0.45$, $a_P = a_W + a_E = 1.0$.

For $u = 2.5 \rightarrow F = 2.5$, $Pe_{cell} = 5 > 2$: central $a_E = 0.5 - 1.25 = -0.75 < 0$ (wiggles).

Upwind: $a_W = 0.5 + 2.5 = 3.0$, $a_E = 0.5$, $a_P = 3.5$ — bounded.

False diffusion

First-order upwind carries a truncation error that looks like extra diffusion $\Gamma_{num} \approx \rho u \Delta x(1 - \nu)/2$ ... for a flow oblique to the grid lines in 2-D $\Gamma_{num} \approx \rho |V| \Delta x \Delta y \sin\theta \cos\theta/(2(\Delta y |\sin\theta| + \Delta x |\cos\theta|))$. It is largest when the flow is at about 45° to the grid and gradients are steep; reduce by refining the grid or using higher-order bounded schemes (QUICK/limiters).

4. Unsteady Problems — Time Integration

1-D unsteady diffusion: $\rho c \partial T/\partial t = \partial/\partial x(k \partial T/\partial x) + S$. Integrate over CV and Δt:

$$\rho c \Delta x (T_P - T_P^0) = \int_t^{t+\Delta t} [(kA/\delta x)_e (T_E - T_P) - (kA/\delta x)_w (T_P - T_W)] \, dt$$

Weighting factor θ: $\int T \, dt = [\theta T + (1 - \theta) T^0] \Delta t$. $\theta = 0$ explicit, ½ Crank–Nicolson, 1 fully implicit.

$$a_P T_P = a_E T_E + a_W T_W + b, \quad a_P = a_E + a_W + a_P^0 \ (\theta = 1)$$

$$a_P^0 = \rho c \Delta x/\Delta t, \quad b = a_P^0 T_P^0 + S \Delta V$$

- **Explicit** ($\theta = 0$): $a_P = a_P^0$. Positive T_P^0 coefficient $a_P^0 - a_E - a_W \geq 0 \rightarrow \Delta t \leq \rho c (\Delta x)^2/2k$ (same as r

$\le \frac{1}{2}$).

- *Fully implicit* ($\theta = 1$): unconditionally stable and bounded, first order in time; used in most CFD codes. Crank–Nicolson: second order but may oscillate for large Δt.

Worked example: 2-D conduction on a 2×2 mesh

Square plate 1 m × 1 m, $k = 1$, no source

Top face $T = 100°C$, other three faces $0°C$. Four CVs ($\Delta x = \Delta y = 0.5$): A, B (top row), C, D (bottom row).

Interior conductance $a = k\Delta y/\Delta x = 1$; boundary (half cell) $a_B = 2k\Delta y/\Delta x = 2$. Cell A: $a_P = 1 + 1 + 2 + 2 = 6$.

$6A = B + C + 2(100) + 2(0)$; $6B = A + D + 200$; $6C = A + D$; $6D = B + C$.

By symmetry $A = B$, $C = D$: $5A - C = 200$, $5C = A$ -> $A = B = 41.7°C$, $C = D = 8.3°C$ (coarse mesh — refine to converge).

Line-by-line TDMA for 2-D

1. Choose a grid line (say row j). Treat φ_W, φ_P, φ_E on this line as unknowns.
2. Move N and S neighbours (latest values) to the right-hand side: $d_i = a_N\varphi_N + a_S\varphi_S + b$.
3. Solve the resulting tridiagonal system by TDMA ($a_i = -a_W$, $b_i = a_P$, $c_i = -a_E$).
4. Sweep the next row; repeat sweeps (alternating row and column directions) until convergence.

Unsteady convection–diffusion (explicit FVM)

$$\varphi_P^{n+1} = \varphi_P^n + (\Delta t/\rho\Delta x)[\,(D_e(\varphi_E - \varphi_P) - D_w(\varphi_P - \varphi_W)) - (F_e\varphi_e - F_w\varphi_w)\,]^n$$

- Stability: upwind explicit requires $\Delta t \le \rho\Delta x/(F + 2D)$ i.e. both the CFL ($\nu\Delta t/\Delta x \le 1$) and the diffusion restriction ($2\alpha\Delta t/\Delta x^2 \le 1$) combined: $\nu + 2r \le 1$.
- Implicit upwind has coefficients $a_P = a_E + a_W + a_P^0 - S_P\Delta x$ — positive and bounded for any Δt.

5. Finite Volume Method via Finite Difference Method

For a uniform Cartesian grid FVM and FDM are closely related. Start from a conservation law and treat the CV as a finite-difference cell with fluxes at half points $i \pm \frac{1}{2}$:

$$\partial u/\partial t + \partial f/\partial x = 0 \quad \text{->} \quad du_i/dt = -(f_{i+1/2} - f_{i-1/2})/\Delta x$$

Here u_i is the cell average $(1/\Delta x)\int u\,dx$ and $f_{i\pm1/2}$ are numerical fluxes at the faces — a finite-difference formula written in flux (conservative) form.

Equivalence with central differencing

- Diffusion: $(\Gamma\varphi_x)_e - (\Gamma\varphi_x)_w$ with linear profiles $= \Gamma(\varphi_E - 2\varphi_P + \varphi_W)/\Delta x$ -> identical to the

3-point FD formula $\Gamma\varphi_{xx}\Delta x$.

- Convection with central face values: $(F/2)(\varphi_E - \varphi_W)$ = FD central difference $\Delta x{\cdot}u_x F$.
- FDM works with point values and Taylor series; FVM works with cell averages and integral balance — FVM remains valid (and conservative) on non-uniform/unstructured grids, whereas FDM expressions need coordinate transformation.

Numerical flux functions — schemes in FVM form

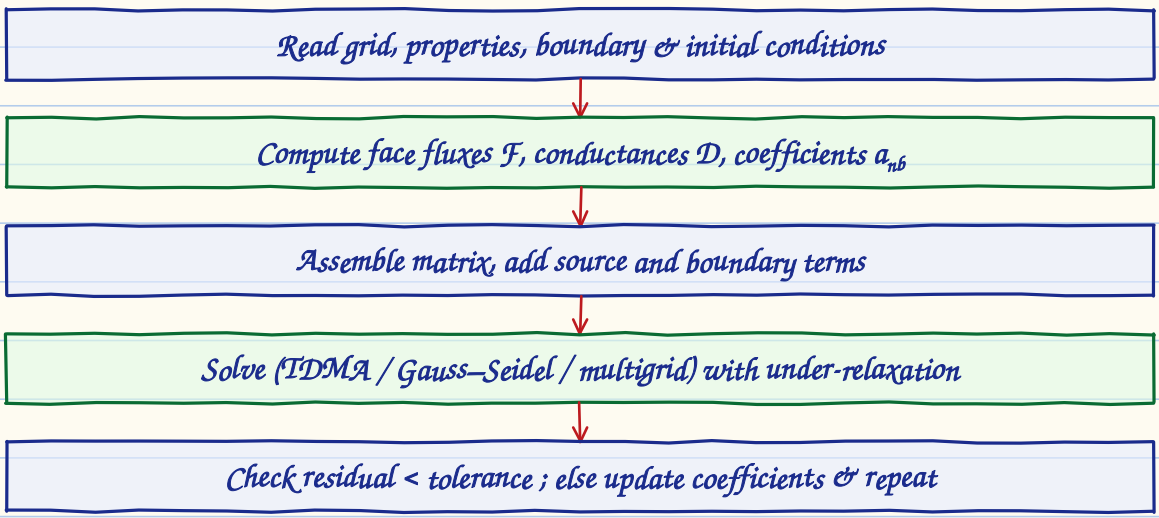
Scheme	Numerical flux $f_{i+1/2}$	Order
Central	$(f_i + f_{i+1})/2$	2
Upwind (a>0)	f_i	1
Lax–Friedrichs	$(f_i + f_{i+1})/2 - (\Delta x/2\Delta t)(u_{i+1} - u_i)$	1
Lax–Wendroff	$(f_i + f_{i+1})/2 - (\Delta t/2\Delta x)A_{i+1/2}(f_{i+1} - f_i)$	2
Godunov	exact Riemann solution at face	1
Roe	$\frac{1}{2}(f_L + f_R) - \frac{1}{2}\lvert A_{Roe}\rvert (u_R - u_L)$	1 (2 with MUSCL)

Update: $u_i^{n+1} = u_i^n - (\Delta t/\Delta x)(f_{i+1/2} - f_{i-1/2})$. Any FD scheme in conservative form can be viewed as an FVM; the choice of the face flux fixes accuracy, dissipation and shock-capturing ability.

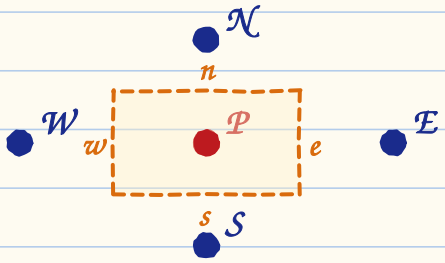
Higher-order reconstruction (MUSCL)

- Reconstruct linear profile inside each cell: $u^L_{i+1/2} = u_i + \frac{1}{2}\psi(r)(u_{i+1} - u_i)$, $u^R_{i+1/2} = u_{i+1} - \frac{1}{2}\psi(r)(u_{i+2} - u_{i+1})$ with limiter ψ -> second-order, TVD, non-oscillatory.

Algorithm of an FVM code



6. Two-dimensional Formulation



Cell P with neighbours W, E (east-west), N, S

faces w, e, n, s (dashed)

$\Delta V = \Delta x \cdot \Delta y \cdot 1$

Steady 2-D convection–diffusion: $\partial(\rho u\varphi)/\partial x + \partial(\rho v\varphi)/\partial y = \partial/\partial x(\Gamma \partial\varphi/\partial x) + \partial/\partial y(\Gamma \partial\varphi/\partial y) + S$.

Integrate over CV ($\Delta x \times \Delta y \times 1$):

$$(J_e - J_w)\Delta y + (J_n - J_s)\Delta x = S\Delta x\Delta y, \quad J = \rho u\varphi - \Gamma\, \partial\varphi/\partial x \ \text{(total flux)}$$

$$F_e = (\rho u)_e \Delta y, \ F_w = (\rho u)_w \Delta y, \ F_n = (\rho v)_n \Delta x, \ F_s = (\rho v)_s \Delta x$$

$$D_e = \Gamma_e \Delta y/\delta x_{PE}, \ D_w = \Gamma_w \Delta y/\delta x_{WP}, \ D_n = \Gamma_n \Delta x/\delta y_{PN}, \ D_s = \Gamma_s \Delta x/\delta y_{SP}$$

$$a_P\varphi_P = a_E\varphi_E + a_W\varphi_W + a_N\varphi_N + a_S\varphi_S + b$$

Coefficient	Upwind / hybrid-type value	Notes
a_E	$D_e + \max(-F_e, 0)$	east
a_W	$D_w + \max(F_w, 0)$	west
a_N	$D_n + \max(-F_n, 0)$	north
a_S	$D_s + \max(F_s, 0)$	south
a_P	$a_E + a_W + a_N + a_S + \Delta F - S_P\Delta x\Delta y$	$\Delta F = 0$ if continuity holds
b	$S_u\Delta x\Delta y$ (+ $a_P^0\varphi_P^0$ unsteady)	source

Pure 2-D diffusion (conduction, uniform grid)

$$a_E = a_W = k\Delta y/\Delta x, \quad a_N = a_S = k\Delta x/\Delta y, \quad a_P = a_E + a_W + a_N + a_S$$

- For $\Delta x = \Delta y = h$ and no source: $T_P = (T_E + T_W + T_N + T_S)/4$ — identical to the five-point finite-difference formula of Unit I.

Solving the 2-D equations

- Pentadiagonal sparse matrix. Gauss–Seidel / SOR point-by-point; or line-by-line TDMA: take unknowns along a line implicitly (tridiagonal) with neighbouring lines (N, S) lagged, sweep in alternating directions (ADI-like).
- Multigrid or conjugate-gradient methods for large grids. Under-relaxation for non-linear problems: $\varphi_P = \varphi_P^{old} + \alpha(\varphi_P^{new} - \varphi_P^{old})$.

- Boundary cells: boundary face fluxes added to source terms (Dirichlet: $a_B = 2\Gamma A/\delta$; Neumann: given flux).

7. Three-dimensional Formulation

Add top (t) and bottom (b) faces for a CV of size $\Delta x \times \Delta y \times \Delta z$. For steady convection–diffusion with continuity satisfied:

$$a_P\varphi_P = a_E\varphi_E + a_W\varphi_W + a_N\varphi_N + a_S\varphi_S + a_T\varphi_T + a_B\varphi_B + b$$

$$a_T = D_t + \max(-F_t, 0), \quad a_B = D_b + \max(F_b, 0), \quad D_t = \Gamma_t \Delta x\Delta y/\delta z_{PT}, \quad F_t = (\rho w)_t \Delta x\Delta y$$

$$a_P = a_E + a_W + a_N + a_S + a_T + a_B - S_P\Delta x\Delta y\Delta z$$

- Seven-diagonal (heptadiagonal) matrix; unsteady fully implicit adds $a_P^0 = \rho\Delta x\Delta y\Delta z/\Delta t$.
- Solvers: Gauss–Seidel with plane/line relaxation (TDMA along lines), ADI with Douglas–Gunn splitting, ICCG / BiCGSTAB, algebraic multigrid.
- Pure diffusion on uniform cube grid: $T_P = (T_E + T_W + T_N + T_S + T_T + T_B)/6$.

Non-orthogonal and unstructured meshes

- Face fluxes use area vectors: $\Sigma_f (\rho V\varphi)_f A_f$ and gradient $(\nabla\varphi)_f A_f = (\varphi_N - \varphi_P)|A_f|/d$·(correction terms for non-orthogonality).
- Gradient by Green–Gauss $(\nabla\varphi)_P = (1/V)\Sigma_f \varphi_f A_f$ or least squares; deferred correction keeps a compact stencil.
- Vertex-centred vs cell-centred storage; volumes may be tetrahedra, hexahedra, prisms, polyhedra.

Extension to the flow equations

- Momentum equations are convection–diffusion equations for u, v, w with pressure-gradient source — pressure-velocity coupling by SIMPLE family (Unit III) on a staggered or collocated grid (Rhie–Chow).
- For compressible flows, FVM of the Euler/NS equations: $\Sigma_{faces}(F\cdot n)_f A_f$ with Roe/AUSM/HLLC flux + MUSCL reconstruction + RK time marching.

Grid types

Grid	Description	Used with
Structured (Cartesian)	regular i,j,k indexing	FDM, simple FVM, ADI/TDMA
Body-fitted curvilinear	mapped to rectangle	FVM with metrics
Unstructured	triangles / tets / polyhedra	FVM, FEM
Hybrid / multi-block	blocks, prism layers	industrial FVM codes

- Grid stretching near walls ($\Delta y_{j+1} = r\Delta y_j$, $r \approx 1.05$-1.2) resolves boundary layers; first cell height from $y^+ \approx 1$ (wall-resolved).
- Cell skewness, aspect ratio, non-orthogonality control accuracy; use smooth variation of cell size.

Verification and grid independence

- Refine grid (e.g. h, $h/2$, $h/4$); observed order $p = \ln[(f_h - f_{h/2})/(f_{h/2} - f_{h/4})]/\ln 2$.
- Richardson extrapolation $f_{exact} \approx f_{h/2} + (f_{h/2} - f_h)/(2^p - 1)$; grid convergence index $GCI = 1.25|\varepsilon|/(r^p - 1)$.
- Validate against experiments/benchmarks (cavity, backward step, flat plate).

Short answer questions

Q1. What is a control volume? -> small cell around node P bounded by faces w, e, n, s over which conservation is enforced.

Q2. Why must $S_P \le 0$? -> to keep $a_P = \Sigma a_{nb} - S_P$ positive: diagonal dominance, stability and boundedness.

Q3. Peclet number? -> $Pe = F/D = \rho u \delta x/\Gamma$, ratio of convective to diffusive strength.

Q4. Why is upwind scheme stable but inaccurate? -> positive coefficients, but first-order false diffusion.

Q5. Difference between FDM and FVM? -> FDM uses Taylor series at points; FVM uses integral balances on cells; conservative on any mesh.

Q6. What is a collocated grid? -> all variables at cell centres; needs Rhie–Chow interpolation to avoid checkerboard pressure.

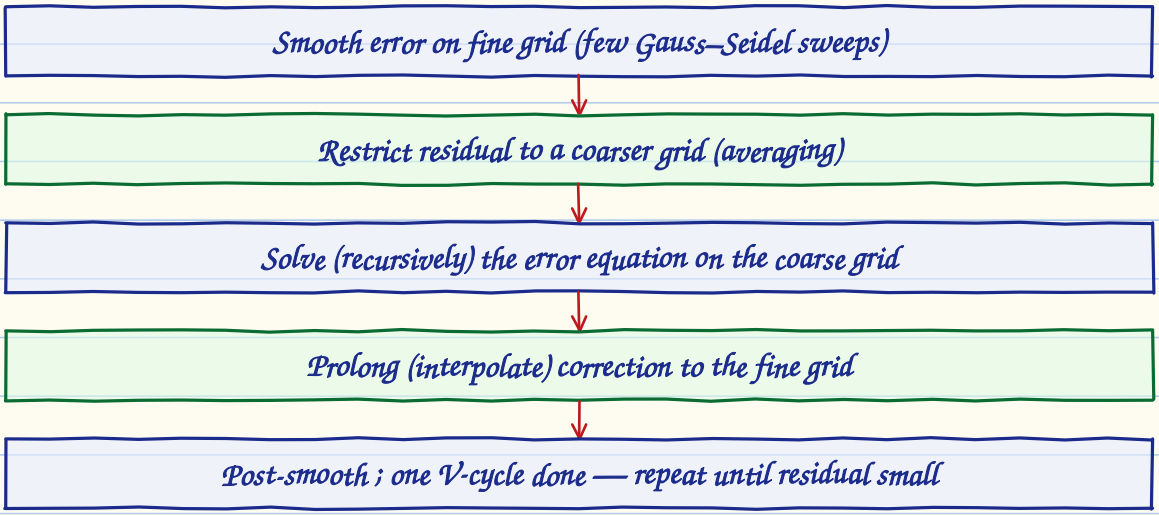
Pressure–velocity coupling in FVM: collocated grid and Rhie–Chow

If u, v, p are all stored at cell centres, a face velocity obtained by simple linear interpolation sees a pressure gradient that ignores neighbouring cells -> checkerboard p. Rhie–Chow adds a pressure-difference smoothing term at each face:

$$u_f = (\bar{u}_P + \bar{u}_E)/2 - d_f[\,(p_E - p_P)/\delta x - \tfrac{1}{2}((\nabla p)_P + (\nabla p)_E)\cdot e_x\,]$$

- The bracket is a fourth-order pressure smoothing term that couples neighbouring pressures; the mass flux built from u_f is used in the continuity (pressure-correction) equation of SIMPLE.

Multigrid acceleration (idea)



- Gauss–Seidel damps high-frequency error quickly but low-frequency error very slowly; coarse grids make low-frequency error look high-frequency. Work ≈ O(N) — much faster than SOR for large 3-D grids.

FVM for the 1-D Euler equations (Riemann-solver fluxes)

Each face has a left state U_L (cell i) and right state U_R (cell i+1): a local Riemann problem. The face flux is taken from its (approximate) solution:

$$U_i^{n+1} = U_i^n - (\Delta t/\Delta x)(F_{i+1/2} - F_{i-1/2}), \qquad F = F(U_L, U_R)$$

- Godunov: exact Riemann solution — accurate for shocks but expensive (iterative).
- Roe: linearised Riemann solver, $F = \tfrac{1}{2}[F(U_L) + F(U_R)] - \tfrac{1}{2}|\tilde{A}|(U_R - U_L)$; needs entropy fix for sonic points.
- HLL / HLLC: use only wave-speed estimates S_L, S_R — robust and simple; AUSM: splits the flux into convective and pressure parts.
- Time step restriction $\Delta t = CFL \cdot \Delta x / \max(|u| + a)$, $CFL \le 1$ for explicit RK/Euler.

Checklist: properties of a good FVM discretisation

Property	Meaning
Consistency	truncation error vanishes as $\Delta x, \Delta t \to 0$
Stability	errors do not amplify (explicit: CFL / diffusion limit)
Convergence	discrete solution → exact solution (Lax equivalence)
Conservativeness	global conservation = sum of local balances
Boundedness	solution stays between boundary / initial extremes
Transportiveness	upstream influence for convection-dominated flow
Accuracy	order, false diffusion, grid refinement behaviour

Axisymmetric and non-Cartesian coordinates

- CV volume and face areas change with r: $\Delta V = r_P \Delta r \Delta\theta \Delta z$, $A_e = r_E \Delta\theta \Delta z$ — coefficients $a_E = \Gamma\, r_P \Delta\theta \Delta z / \delta r$ etc.; metrics are absorbed in areas and volume, so the 1-D form $a_P \varphi_P = \Sigma a_{nb} \varphi_{nb} + b$ remains unchanged.

8. Summary and Important Questions

Quick revision — Unit IV

- FVM: integrate over CV -> flux balance -> $a_P \varphi_P = \Sigma a_{nb} \varphi_{nb} + b$.
- Diffusion: $a_{nb} = \Gamma A / \delta x$; convection–diffusion: central needs $Pe < 2$; upwind bounded but diffusive; hybrid/power-law/QUICK.
- FVM via FDM: $du_i/dt = -(f_{i+1/2} - f_{i-1/2})/\Delta x$; choose numerical flux.
- 2-D: five coefficients $a_{E,W,N,S,P}$; 3-D: add a_T, a_B (seven-point stencil).
- Solvers: TDMA line-by-line, Gauss–Seidel, multigrid.

1. Explain the finite volume method and derive the discretised equation for 1-D steady diffusion with boundary conditions.
2. Derive the discretisation of 1-D convection–diffusion by central, upwind and hybrid schemes; discuss Peclet number.
3. Show how the finite volume method is related to the finite difference method.
4. Formulate the FVM for 2-D and 3-D steady convection–diffusion problems.
5. Explain the implicit time-marching FVM for unsteady diffusion and its boundedness.

APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – V

Standard Variational Methods

Syllabus (as given)

- Standard variational methods: linear fluid flow problems, steady state problems, transient problems.

1. Introduction — Variational Principle and Weak Form

Variational methods obtain the approximate solution of a differential equation by (i) minimising an integral quantity called a functional, or (ii) forcing the weighted integral of the residual to zero. They are the mathematical foundation of the finite element method (FEM).

Functional and Euler–Lagrange equation

A functional $I[u]$ assigns a number to a function $u(x)$. Example (1-D Poisson / heat conduction / Poiseuille flow):

$$I[u] = \int_0^L [\tfrac{1}{2} k (du/dx)^2 - f u] \, dx$$

Among all admissible functions (satisfying essential BCs), the true solution makes I stationary: $\delta I = 0$

-> Euler–Lagrange equation:

$$-d/dx(k \, du/dx) = f \qquad \text{(differential equation)}$$

- Essential (Dirichlet) BCs: $u = \bar{u}$ on boundary — must be satisfied by the trial function.
- Natural (Neumann) BCs: $k \, du/dx = q$ — arise automatically from the variation; need not be imposed on the trial function.
- For self-adjoint (symmetric) operators the functional exists and gives symmetric stiffness matrices; for fluid flow with convection (non-self-adjoint) we use Galerkin weighted residuals instead.

Analogy

- Structural mechanics: functional = total potential energy; minimum potential energy principle.
- Heat conduction / Stokes flow: functional = dissipation-type integral; Stokes flow minimises viscous dissipation (Helmholtz minimum dissipation theorem).

2. Rayleigh–Ritz Method

Assume a trial solution as a finite series of known basis functions:

$$u(x) \approx u_h(x) = \sum_{j=1}^{n} c_j \varphi_j(x) \quad (\varphi_j \text{ satisfy the essential BCs})$$

Substitute in the functional $I(c_1, \ldots, c_n)$ and require stationarity: $\partial I/\partial c_i = 0$, $i = 1 \ldots n$ -> matrix equation

$$\sum_j K_{ij} c_j = F_i, \quad K_{ij} = \int k \varphi_i' \varphi_j' \, dx, \quad F_i = \int f \varphi_i \, dx$$

Example — Ritz solution

$-u'' = 1$ on $0 < x < 1$, $u(0) = u(1) = 0$. Trial $u = c\,x(1 - x)$ (satisfies BCs).

$I = \int [\tfrac{1}{2}(c(1 - 2x))^2 - c\,x(1 - x)]\, dx = c^2/6 - c/6$ -> $dI/dc = c/3 - 1/6 = 0$ -> $c = 1/2$.

$u = x(1 - x)/2$ which is the exact solution (the trial space contained it).

3. Method of Weighted Residuals (MWR)

When no functional exists (e.g. convection, Navier–Stokes), use the residual of the differential equation $L(u) = f$:

$$R(x) = L(u_h) - f \neq 0 \quad \text{->} \quad \int_\Omega W_i(x) R(x)\, dx = 0, \quad i = 1 \ldots n$$

Method	Weight function W_i
Collocation	Dirac delta $\delta(x - x_i)$ ($R = 0$ at chosen points)
Subdomain	1 on subdomain Ω_i, 0 elsewhere (average $R = 0$)
Least squares	$W_i = \partial R/\partial c_i$ (minimise $\int R^2\, dx$)
Moments	$W_i = x^{i-1}$
Galerkin	$W_i = \varphi_i$ (same as basis/shape function)

- Galerkin method gives symmetric matrices for self-adjoint operators and, in that case, equals the Ritz method; it is the standard choice in FEM.

Weak (variational) form of the 1-D problem

Multiply $-(k u')' = f$ by test function w ($w = 0$ at essential BC) and integrate by parts:

$$\int_0^L k w' u' \, dx = \int_0^L f w \, dx + [\, w k u' \,]_0^L$$

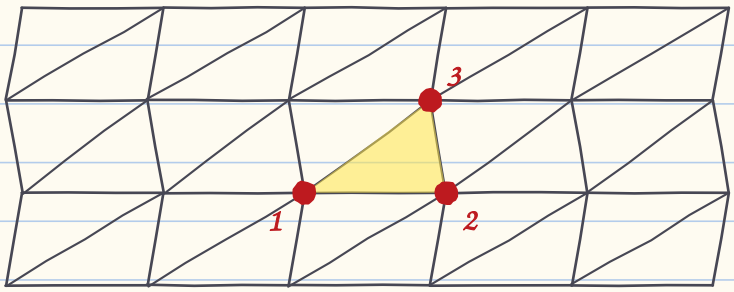
- Integration by parts lowers the derivative order (C^0 shape functions suffice) and brings in natural BCs through the boundary term.

Galerkin example

Same problem $-u'' = 1$, trial $u = a\,x(1-x)$, $w = x(1-x)$: $\int (u'w' - w)dx = 0$.

$\int (a(1-2x)^2 - x(1-x))dx = a/3 - 1/6 = 0 \Rightarrow a = 1/2$. √

4. Finite Element Method — Discretisation

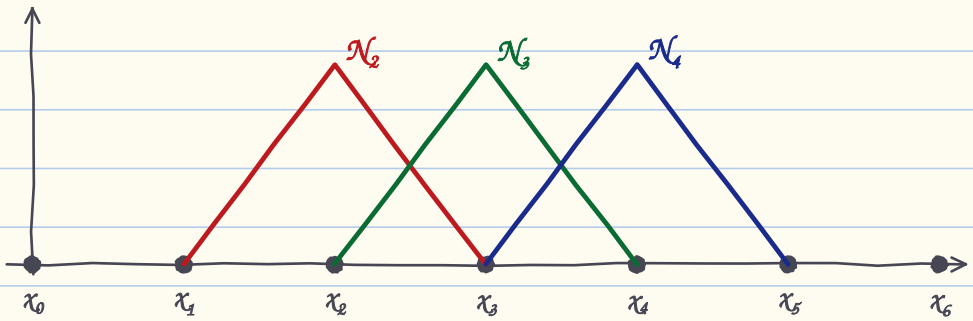


Element e (nodes 1,2,3)

Nodes = unknowns u_j

Mesh = assembly of elements

1. Divide the domain into elements connected at nodes (1-D: lines; 2-D: triangles/quads; 3-D: tets/hexes).
2. Choose local shape (interpolation) functions $N_j(x)$: $u^e(x) = \Sigma\, N_j^e(x)\, u_j$.
3. Apply Galerkin weighted residual / variational statement to every element → element matrices.
4. Assemble element matrices into the global system; impose boundary conditions; solve $K u = F$.
5. Post-process: gradients, fluxes, velocities, error estimates.



$N_i(x_i) = 1$

$N_i(x_j) = 0$

$(i \neq j)$

- Global "hat" functions: $N_i = 1$ at node i, 0 at all other nodes — compact support, so the matrix is sparse and banded.

1-D linear element (length $h = x_2 - x_1$)

$$N_1 = (x_2 - x)/h\,, \quad N_2 = (x - x_1)/h \qquad (N_1 + N_2 = 1)$$

$$K^e = (k/h)\,[\,1\ \ -1\,;\,-1\ \ 1\,]\,, \qquad F^e = (f\,h/2)\,[\,1\,;\,1\,]\,, \qquad M^e = (\rho\,h/6)\,[\,2\ \ 1\,;\,1\ \ 2\,]$$

$K^e_{ij} = \int k N_i' N_j'\, dx$, $F^e_i = \int f N_i\, dx$, $M^e_{ij} = \int \rho N_i N_j\, dx$ (consistent mass matrix).

2-D linear triangular element

$$N_i = (a_i + b_i x + c_i y)/(2A), \quad b_1 = y_2 - y_3,\ c_1 = x_3 - x_2 \text{ (cyclic)}$$

$$K^e_{ij} = (k/4A)(b_i b_j + c_i c_j), \quad F^e_i = fA/3, \quad M^e_{ij} = (\rho A/12)(1 + \delta_{ij})$$

A = area of triangle. Quadrilateral elements: isoparametric mapping with natural coordinates (ξ, η) and Gauss quadrature $\int f\, d\xi \approx \Sigma\, w_g f(\xi_g)$.

Quadratic element and natural coordinates (1-D)

Use $\xi \in [-1, 1]$ with $x = \Sigma\, N_i(\xi) x_i$ (isoparametric). Jacobian $J = dx/d\xi = h/2$ for a straight element.

$$\text{Linear: } N_1 = (1 - \xi)/2, \quad N_2 = (1 + \xi)/2$$

$$\text{Quadratic (3 nodes at } \xi = -1, 0, 1\text{): } N_1 = \xi(\xi - 1)/2, \quad N_2 = 1 - \xi^2, \quad N_3 = \xi(\xi + 1)/2$$

$$K^e_{ij} = \int_{-1}^{1} k\,(dN_i/d\xi)(dN_j/d\xi)(1/J)\, d\xi, \quad F^e_i = \int_{-1}^{1} f N_i J\, d\xi$$

Gauss points n	Locations ξ_g	Weights w_g
1	0	2
2	±0.57735	1, 1
3	0, ±0.77460	8/9, 5/9, 5/9

- n-point Gauss quadrature is exact for polynomials of degree $2n - 1$. Quadratic stiffness matrix: $K^e = (k/3h)[7\ -8\ 1;\ -8\ 16\ -8;\ 1\ -8\ 7]$.
- Quadratic elements give $O(h^3)$ in L^2 and $O(h^2)$ in energy norm — much higher accuracy for the same number of nodes.

5. Linear Fluid Flow Problems

"Linear" = governing equations are linear in the unknowns: potential flow, creeping (Stokes) flow, fully developed laminar duct flow, unsteady diffusion of momentum/vorticity — no convective $(u \cdot \nabla u)$ term.

5.1 Potential (irrotational, incompressible) flow

$$\nabla^2 \varphi = 0 \text{ in } \Omega, \quad u = \partial\varphi/\partial x,\ v = \partial\varphi/\partial y; \quad \varphi = \varphi_0\ (\Gamma_1), \quad \partial\varphi/\partial n = u_n\ (\Gamma_2)$$

$$I[\varphi] = \tfrac{1}{2} \int_\Omega (\varphi_x^2 + \varphi_y^2)\, d\Omega - \int_{\Gamma_2} u_n \varphi\, d\Gamma$$

- Stream function version $\nabla^2 \psi = 0$ (ψ = const on solid walls). Element equations are those of Laplace/Poisson: $K_{ij} = \int \nabla N_i \cdot \nabla N_j d\Omega$.

5.2 Fully developed laminar flow in a duct (Poiseuille / Poisson problem)

$$\mu \nabla^2 w = dp/dz = -G, \quad w = 0 \text{ on walls}$$

$$I[w] = \int_\Omega [\, \tfrac{1}{2}\mu (w_x^2 + w_y^2) - G\, w \,]\, d\Omega$$

Same Poisson form as heat conduction with heat source: $K_{ij} = \mu \int \nabla N_i \cdot \nabla N_j d\Omega$, $F_i = G \int N_i d\Omega$.

Volume flow rate $Q = \int w\, d\Omega$; friction factor from the wall gradient.

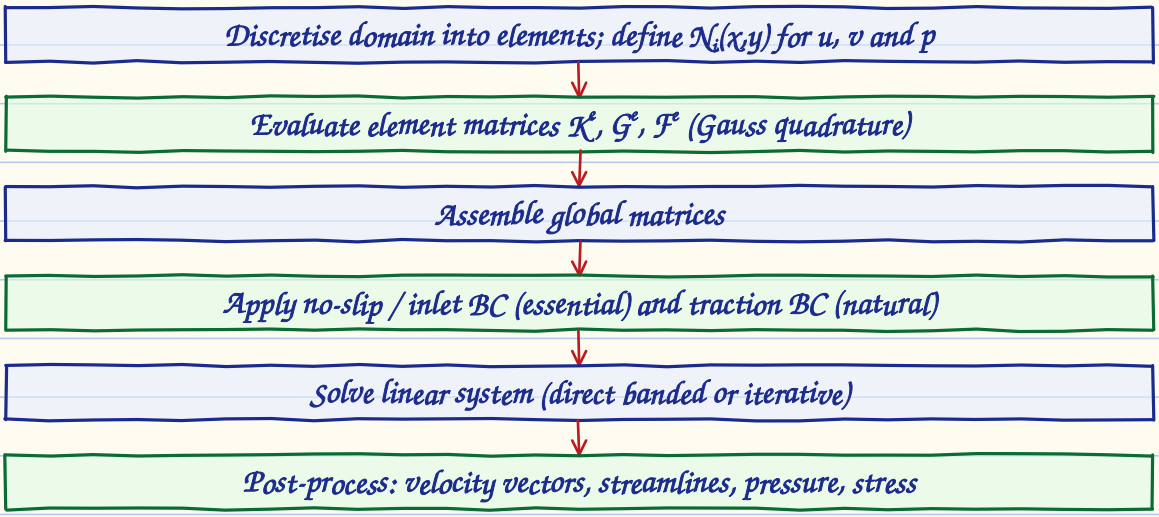
5.3 Stokes (creeping) flow — mixed formulation

$$-\mu \nabla^2 \mathcal{V} + \nabla p = f, \quad \nabla \cdot \mathcal{V} = 0$$

Weak form: $\int \mu\, \nabla w : \nabla \mathcal{V}\, d\Omega - \int p\, \nabla \cdot w\, d\Omega = \int w \cdot f\, d\Omega$; $\int q\, \nabla \cdot \mathcal{V}\, d\Omega = 0$

Matrix system: $[\, K \;\; -G^T \;;\; G \;\; 0 \,]\, [\, \mathcal{V} \;;\; P \,] = [\, F \;;\; 0 \,]$

- **LBB (Babuška–Brezzi) condition:** velocity and pressure spaces cannot be chosen arbitrarily (equal-order linear elements give spurious pressure modes). Stable pairs: Taylor–Hood P2–P1 (quadratic velocity, linear pressure), Q2–Q1, MINI element.
- **Penalty method:** relax incompressibility $p = -\lambda\, \nabla \cdot \mathcal{V}$ ($\lambda \sim 10^6\, \mu$) and eliminate p: $(K + \lambda\, G^T M_p^{-1} G)\mathcal{V} = F$ — reduced integration of the penalty term avoids locking.
- **Stream function–vorticity:** biharmonic $\nabla^4 \psi = 0$ needs C^1 elements; or split into two Poisson equations.



Applying boundary conditions to $K\,u = F$

Method	How
Elimination	strike out rows/columns of known u_k; move $K_{ik} u_k$ to the right side
Penalty	$K_{kk} \leftarrow K_{kk} + C$, $F_k \leftarrow F_k + C\, \bar{u}_k$ with $C \sim 10^6 \max K_{ii}$
Lagrange multiplier	add constraint equation $u_k = \bar{u}_k$; enlarges the system
Natural BC	add boundary load q (or $h(T_\infty - T)$) to F_i and h to K_{ii} (Robin)

Worked example: 2-D triangle element

Element matrix for Laplace / Poisson (k = 1)

Triangle with nodes 1 (0,0), 2 (1,0), 3 (0,1): $A = 1/2$. $b_1 = y_2 - y_3 = -1$, $b_2 = y_3 - y_1 = 1$, $b_3 = y_1 - y_2 = 0$;

$c_1 = x_3 - x_2 = -1$, $c_2 = x_1 - x_3 = 0$, $c_3 = x_2 - x_1 = 1$.

$K_{11} = (b_1^2 + c_1^2)/4A = 2/2 = 1$, $K_{12} = (b_1b_2 + c_1c_2)/4A = -1/2$, $K_{13} = -1/2$, $K_{22} = 1/2$, $K_{23} = 0$, $K_{33} = 1/2$.

Row sums are zero (constant field gives zero gradient)√; matrix symmetric √.

Assembly of three 1-D elements (non-uniform k)

Assembly rule

Elements e1 (nodes 1-2, k_1, h_1), e2 (2-3), e3 (3-4). Global K (4×4) is tridiagonal:

$K_{11} = k_1/h_1$, $K_{22} = k_1/h_1 + k_2/h_2$, $K_{33} = k_2/h_2 + k_3/h_3$, $K_{44} = k_3/h_3$, $K_{12} = -k_1/h_1$, …

Each element contributes to the diagonal of its two nodes (adds) and to the off-diagonal between them — interfaces between materials / fluids with different μ are handled naturally.

6. Steady State Problems

Steady problems give a system $Ku = F$ with no time derivative. Worked 1-D example (axial heat conduction or Poiseuille flow between plates):

Example: $-u'' = 1$, $u(0) = u(1) = 0$, two linear elements ($h = 0.5$)

Nodes $x = 0, 0.5, 1$. Element matrices $K^e = (1/0.5)[1\ -1; -1\ 1] = [2\ -2; -2\ 2]$, $F^e = (0.5/2)(1, 1) = (0.25, 0.25)$.

Assembly: $K = [2\ -2\ 0; -2\ 4\ -2; 0\ -2\ 2]$, $F = (0.25, 0.5, 0.25)$.

Apply $u_1 = u_3 = 0$ (delete rows/columns 1, 3): $4u_2 = 0.5$ -> $u_2 = 0.125$.

Exact $u(0.5) = x(1 - x)/2 = 0.125$ √ (nodally exact for this 1-D problem).

Plane Poiseuille flow by FEM

Flow between plates at $y = \pm b$ with pressure gradient $G = -dp/dx$: $\mu w'' = -G$. With symmetry, solve on $0 \le y \le b$ with $w(b) = 0$ (essential) and $w'(0) = 0$ (natural, symmetric). Exact: $w = (G/2\mu)(b^2 - y^2)$, $w_{max} = Gb^2/2\mu$, mean velocity $= 2w_{max}/3$.

Steps for steady-state fluid problems

1. Weak form from governing equation (Galerkin).
2. Element matrices (shape functions, numerical integration).
3. Assembly: $K_{IJ} \mathrel{+}= K^{e}_{ij}$ where I, J = global numbers of local nodes i, j.
4. Essential BC: delete rows/columns, or penalty (large number on diagonal), or modify RHS.
5. Solve; recover derived variables (shear stress $\tau = \mu\, du/dy$, flux) — they are less accurate than u (one order lower).

- Convection-dominated steady flows (non-self-adjoint) show wiggles with plain Galerkin for large Peclet number -> use SUPG (streamline upwind Petrov–Galerkin): $W_i = N_i + (\alpha h/2\,|u|)u\cdot\nabla N_i$.
- Non-linear steady problems (Navier–Stokes): Picard (successive substitution) or Newton–Raphson iterations on $K(u)u = F$.

Convection–diffusion by Galerkin / SUPG (steady, 1-D)

$$u\, d\varphi/dx = \Gamma\, d^2\varphi/dx^2 \;\rightarrow\; \int (W u \varphi' + \Gamma W \varphi')\, dx = 0$$

Linear elements, uniform mesh, Galerkin gives at an interior node:

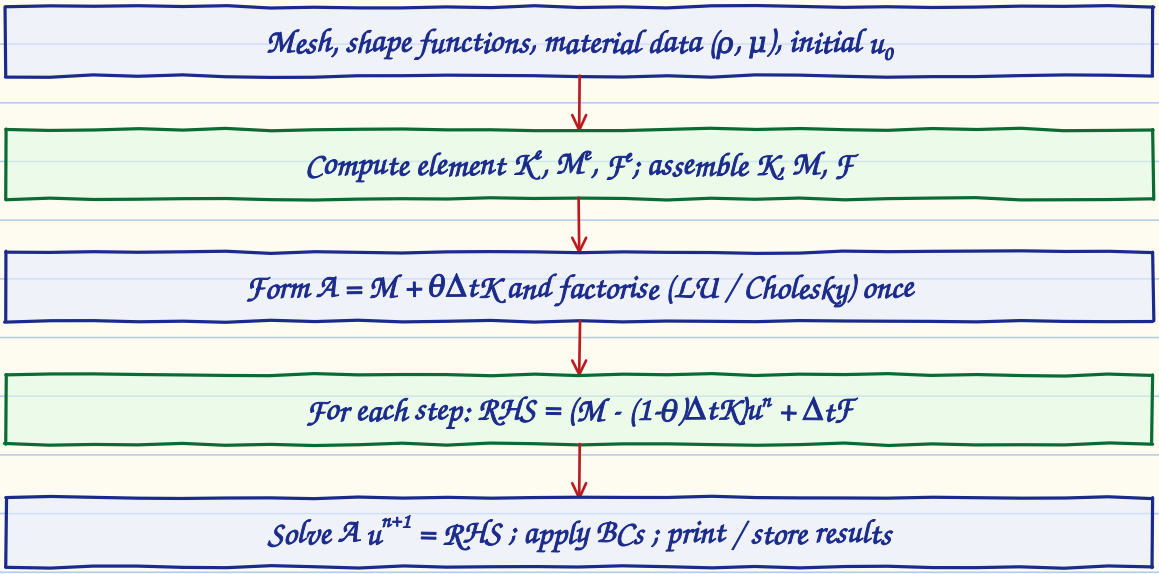
$(u/2)(\varphi_{i+1} - \varphi_{i-1}) = (\Gamma/h)(\varphi_{i+1} - 2\varphi_i + \varphi_{i-1})$ — identical to central differencing -> wiggles for $Pe_h = uh/2\Gamma > 1$.

- SUPG adds artificial diffusion only along the streamline: $\Gamma_{art} = \xi(Pe)\, u\, h/2$, $\xi = \coth(Pe) - 1/Pe$ (optimal 1-D value, gives nodally exact solution).
- Other stabilisations: GLS (Galerkin least squares), bubble functions, Taylor–Galerkin (time-dependent), characteristic Galerkin.

Pipe/duct flow and Couette flow examples

- Couette flow between plates (gap H, top plate speed U): $\mu w'' = 0$ with $w(0) = 0$, $w(H) = U$ -> linear profile; one linear element reproduces it exactly.
- Hagen–Poiseuille flow in a circular pipe: axisymmetric weak form $\int \mu\, (dN_i/dr)(dw/dr)\, r\, dr = \int G\, N_i\, r\, dr$; exact $w = (G/4\mu)(R^2 - r^2)$, $Q = \pi G R^4/8\mu$.
- Flow in a rectangular duct: 2-D Poisson on the cross-section, 4 triangular or quadrilateral elements per quadrant already gives Q within a few percent.
- Potential flow around a cylinder in a channel: solve $\nabla^2\psi = 0$ with $\psi = 0$ on the body and the free-stream $\psi = U y$ far away; pressure from Bernoulli $p = p_\infty + \tfrac{1}{2}\rho(U^2 - |\nabla\psi|^2)$.

Algorithm for a transient FEM flow solver



7. Transient Problems

Time-dependent linear problem, e.g. unsteady diffusion of momentum (Stokes first problem) $\rho\,\partial u/\partial t = \mu\,\partial^2 u/\partial y^2$ or heat conduction. Galerkin in space (semi-discretisation):

$$M\dot{u} + Ku = F(t) \qquad \text{(system of ODEs)}$$

where $M_{ij} = \int \rho N_i N_j\,dx$ (mass matrix), $K_{ij} = \int \mu\, N_i' N_j'\,dx$. Initial condition $u(0) = u_0$.

Time integration — θ-method (generalised trapezoidal)

$$(M + \theta\Delta t\,K)\,u^{n+1} = (M - (1-\theta)\Delta t\,K)\,u^n + \Delta t[\,\theta F^{n+1} + (1-\theta)F^n\,]$$

θ	Name	Stability	Accuracy
0	Forward Euler	$\Delta t \le 2/\lambda_{max}$	$O(\Delta t)$
½	Crank–Nicolson	unconditional	$O(\Delta t^2)$
2/3	Galerkin	unconditional	$O(\Delta t)$
1	Backward Euler	unconditional	$O(\Delta t)$

- λ_{max} = largest eigenvalue of $M^{-1}K \sim 4\mu/(\rho h^2)$ for linear elements: explicit limit $\Delta t \le \rho h^2/(2\mu)$ (for lumped mass $\Delta t \le \rho h^2/2\mu$).
- For $\theta \ge ½$ unconditionally stable; for $\theta < ½$ conditionally stable: $\Delta t \le 2/((1-2\theta)\lambda_{max})$.
- Lumped mass matrix $M_L = (\rho h/2)I$ (row sum) makes explicit schemes cheap (diagonal M); consistent M is more accurate for wave problems but can give oscillations with $\theta = 0$.
- Each step solves a linear system with the same matrix $(M + \theta\Delta t K)$ — factorise once, reuse; this is the great advantage for linear problems.

Alternatives

- Finite difference in time: Backward Euler / BDF2; Newmark-β for second-order systems $M\ddot{u} + C\dot{u} + Ku = F$.
- Space–time elements, discontinuous Galerkin in time; modal (eigenfunction) expansion $u = \Sigma a_k(t)\psi_k(x)$ giving uncoupled ODEs $\dot{a}_k + \lambda_k a_k = 0$.

Example: transient 1-D heat/diffusion with one free node

Rod $0 \le x \le 1$, two elements ($h = 0.5$), $u(0) = u(1) = 0$, $\rho = \mu = 1$, no source, $u(0.5, 0) = 1$.

Free node 2: $M_{22} = 2h/3 = 1/3$, $K_{22} = 4$ -> $(1/3)\dot{u} + 4u = 0$ -> $\dot{u} = -12u$, exact semi-discrete $u = e^{-12t}$ (true: $e^{-\pi^2 t} \approx e^{-9.87t}$).

Backward Euler: $u^{n+1} = u^n/(1 + 12\Delta t)$; Crank–Nicolson: $u^{n+1} = u^n(1 - 6\Delta t)/(1 + 6\Delta t)$;

Forward Euler: stable only if $12\Delta t \le 2$ -> $\Delta t \le 1/6$.

Transient fluid-flow applications

- Stokes first/second problem (impulsively moved plate, oscillating plate): diffusion of vorticity; boundary-layer thickness $\delta \sim \sqrt{(\nu t)}$.
- Start-up of Poiseuille flow in a channel/pipe: $u(y,t) \to$ steady parabola as $t \to \infty$; time scale $t \sim b^2/\nu$.
- Unsteady Stokes with incompressibility: saddle-point system solved at each time step by projection / mixed elements.
- Unsteady Navier–Stokes (non-linear): operator splitting, characteristic-Galerkin or SUPG for convection, Newton iterations per step.

Short answer questions

Q1. Difference between Ritz and Galerkin? -> Ritz needs a functional (self-adjoint problem); Galerkin works directly with the differential equation.

Q2. What is a shape function? -> interpolation function with $N_i = 1$ at node i, 0 at other nodes; $\Sigma N_i = 1$.

Q3. What are essential and natural BCs? -> essential: imposed on u; natural: appear in the weak form (fluxes/tractions).

Q4. Why integrate by parts? -> reduces required continuity (C^0) and introduces natural BCs.

Q5. What is the consistent vs lumped mass matrix? -> consistent: $\int \rho N_i N_j$; lumped: diagonal (row sum), used for explicit schemes.

Q6. Why Galerkin is poor for high Peclet flows? -> central-like, oscillations; use SUPG.

Q7. What is locking in penalty methods? -> too large λ makes the matrix singular, solved by reduced/selective integration.

Comparison: Ritz vs Galerkin vs collocation (example $-u'' = 1$)

Method	Weight / condition	Result with $u = a\,x(1 - x)$
Ritz	$\partial I/\partial a = 0$	$a = 1/2$
Galerkin	$W = x(1 - x)$	$a = 1/2$
Collocation at $x = ½$	$R(½) = 0$	$2a - 1 = 0$ -> $a = 1/2$
Least squares	$\int R^2$ min	$a = 1/2$

All methods give the exact solution here because the trial function belongs to the solution space; in general they differ.

7A. Beyond the Linear Case — Weak Form of Navier–Stokes (brief)

$$\int w\cdot(\rho\partial V/\partial t + \rho(V\cdot\nabla)V)\,d\Omega + \int \mu\,\nabla w:\nabla V\,d\Omega - \int p\,\nabla\cdot w\,d\Omega = \int w\cdot f\,d\Omega + \int_\Gamma w\cdot t\,d\Gamma, \quad \int q\,\nabla\cdot V\,d\Omega = 0$$

- Extra non-linear convective integral gives a non-symmetric, solution-dependent matrix $K(V)$; solve with Picard or Newton iteration (linearise $V\cdot\nabla V \approx V^k\cdot\nabla V^{k+1}$).
- Penalty / segregated (projection) / coupled solvers; time step limited by CFL for explicit convection.
- Vorticity–stream function FEM: two Poisson problems $\int\nabla N\cdot\nabla\psi\,d\Omega = \int N\omega\,d\Omega$ and vorticity transport solved with SUPG; wall vorticity from natural BC of ψ.

Element	Nodes / DOF	Use in fluid problems
2-node line	linear	1-D channel, pipe network
3-node triangle (T3)	linear	Laplace, Poisson, ψ, φ
6-node triangle (T6)	quadratic	velocity in Taylor–Hood
4-node quad (Q4)	bilinear	pressure, temperature
8/9-node quad	biquadratic	velocity, high accuracy
4-node tetrahedron	linear	3-D potential flow

Properties of the FEM matrices

- K is symmetric positive definite for self-adjoint problems with essential BCs; sparse banded -> Cholesky or conjugate-gradient solvers.
- Stokes saddle-point matrix is symmetric indefinite; solved with Uzawa iteration, Schur

complement or block preconditioned GMRES/MINRES.

- Bandwidth minimisation by good node numbering (Cuthill–McKee) reduces memory and solve time.

8. Error Estimates and Convergence

- Energy-norm error for elements of degree p: $||u - u_h||_E \le C h^p |u|_{p+1}$ and in L^2 norm $O(h^{p+1})$ (linear elements: 1st order in energy norm, 2nd order in L^2).
- Ritz/Galerkin solution is the best approximation to the exact solution in the energy norm (Céa's lemma).
- h-refinement (smaller elements), p-refinement (higher order), r-refinement (node movement), adaptive meshing with a-posteriori error estimators.

Aspect	FEM (variational)	FVM / FDM
Basic idea	minimise functional / weak form	flux balance / Taylor series
Geometry	excellent, unstructured	good (FVM) / structured (FDM)
BCs	natural BC built-in	imposed through fluxes/ghost nodes
Conservation	global weak sense	local exact (FVM)

Quick revision — Unit V

- Functional $I[u] = \int (\tfrac{1}{2} k u'^2 - f u)dx$; stationarity gives the differential equation + natural BC.
- Ritz: minimise I over trial space; Galerkin: weight function = shape function (works for any PDE).
- Element: $K^e = (k/h)[1\ -1;\ -1\ 1]$, $F^e = (fh/2)(1, 1)$, $M^e = (\rho h/6)[2\ 1;\ 1\ 2]$.
- Steady: $Ku = F$; Transient: $M\dot{u} + Ku = F$, θ-scheme stable unconditionally for $\theta \ge \tfrac{1}{2}$.
- Stokes flow: mixed (V, p) system needs LBB-stable elements (Taylor–Hood) or penalty method.

Important Questions

1. What is a functional? Derive the Euler–Lagrange equation for $I = \int (\tfrac{1}{2} k u'^2 - f u)dx$.
2. Explain Rayleigh–Ritz and Galerkin methods with an example; compare MWR techniques.
3. Derive element stiffness matrix for a 1-D linear element; assemble and solve a 2-element problem.

4. Formulate the Galerkin FEM for Stokes flow / potential flow; explain LBB condition.
5. Derive $M\dot{u} + Ku = F$ for transient diffusion and discuss the θ-method and its stability.